Algebra

Applications and Strategies

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Preliminary Edition

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Acknowledgements

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Preface

Hello to "you" the student. As this course has evolved to support our computer science major, so has the text. The current version of the book is intended to support inquiry-based teaching for understanding that is so crucial for future teachers, while also providing the necessary mathematical foundation and application-based motivation for computer science students. While teaching the course in Spring 2024 using an early version of this edition, I was pleasantly surprised by how many students reported that they, for the first time, saw how useful math could be in the "real world." I hope that this experience can be replicated in other classes using this text.

This book is intended to be used in a class taught using problem-oriented or inquiry-based methods. Each section begins with a preview of the content that includes an open-ended *Investigate!* motivating question, as well as a structured preview activity. The preview activities are carefully scaffolded to provide an entry-point to the section's topic and to prime students to engage deeply in the material. Depending on the pace of the class, I have found success assigning only the section preview before class, using the preview activity as in-class group work, or assigning the entire section to be read before class (each section concludes with a small set of reading questions that can be assigned to encourage students to actually read). For those readers using this book for self-study, the organization of the sections will hopefully mimic the style of a rich inquiry-based classroom.

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How to Use This Book

In addition to expository text, this book has a few features designed to encourage you to interact with the mathematics.

***Investigate!* questions.** Sprinkled throughout the sections (usually at the very beginning of a topic) you will find open-ended questions designed to engage you with the topic soon to be discussed. You really should spend some time thinking about, or even working through, these problems before reading the section. However, don't worry if you cannot find a satisfying solution right away. The goal is to pique your interest, so you will read what is next looking for answers.

Preview Activities. Most sections include a structured preview activity. These contain leading questions that you should be able to completely answer before reading the section. The idea is that the questions prime you to engage meaningfully with the new content ahead. If you are using the online version, most of these questions will provide you with immediate feedback so you can be confident moving forward.

Examples. I have tried to include the "correct" number of examples. For those examples that include *problems*, full solutions are included. Before reading the solution, try to at least have an understanding of what the problem is asking. Unlike some textbooks, the examples are not meant to be all-inclusive for problems you will see in the exercises. They should not be used as a blueprint for solving other problems. Instead, use the examples to deepen your understanding of the concepts and techniques discussed in each section. Then use this understanding to solve the exercises at the end of each section.

Exercises. You get good at math through practice. Each section concludes with practice problems meant to solidify concepts and basic skills presented in that section; the online version provides immediate feedback on these problems. There are then additional exercises that are more challenging and open-ended. These might be assigned as written homework or used in class as group work. Some of the additional exercises have hints or solutions in the back of the book, but use these as little as possible. Struggle is good for you. At the end of each chapter, a larger collection of similar exercises is included (as a sort of "chapter review") which might bridge the material of different sections in that chapter.

Interactive Online Version. For those of you reading this in print or as a PDF, I encourage you to also check out the interactive online version. Many of the preview activities and exercises are interactive and can give you immediate feedback. Some of these have randomized components, allowing you to practice many similar versions of the same problems until you master the topic.

Hints and solutions to examples are also hidden away behind an extra click to encourage you to think about the problem before reading the solution. There is a good search feature available as well, and the index has expandable links to see the content without jumping to the page immediately. There is also a python scratch pad (the pencil icon) so you can try out some code if you feel so inclined.

Additional interactivity is planned. These "bonus" features will be added on a rolling basis, so keep an eye out!

You can view the interactive version for free at `discrete.openmathbooks.org` or by scanning the QR code below.

Contents

Chapter 0

Introduction

Welcome to this algebra class!

0.1 What is Discrete Mathematics?

dis·crete / dis'krēt.

Adjective: Individually separate and distinct.

Synonyms: separate - detached - distinct - abstract.

Defining *discrete mathematics* is hard because defining *mathematics* is hard. What is mathematics? The study of numbers? In part yes, but you also study functions and lines and triangles and parallelepipeds and vectors and . . . . Or perhaps you want to say that mathematics is a collection of tools that allow you to solve problems. What sort of problems? Well, those that involve numbers, functions, lines, triangles, ... Whatever your conception of what mathematics is, try applying the concept of "discrete" to it, as defined above. Some math fundamentally deals with *stuff* that is individually separate and distinct.

In an algebra or calculus class, you might have found a particular set of numbers (perhaps they constitute the range of a function). You would represent this set as an interval: $[0, \infty)$ is the range of $f(x) = x^2$ since the set of outputs of the function are all real numbers 0 and greater. This set of numbers is NOT discrete. The numbers in the set are not separated by much at all. In fact, take any two numbers in the set and there are infinitely many more between them that are also in the set.

Discrete math could still ask about the range of a function, but the set would not be an interval. Consider the function that gives the number of children of each person reading this. What is the range? I'm guessing it is something like $\{0, 1, 2, 3, 4\}$. Maybe 5 or 6 is in there too.[1] But certainly nobody reading this has 1.32419 children. This output set *is* discrete because the elements are separate. The inputs to the function also form a discrete set because each input is an individual person.

There are many discrete mathematical objects besides sets of numbers; we will introduce some of these in REMOVED reference. Studying these discrete **structures** is the main focus of discrete mathematics and this book. However, the reason we want to study these structures is because they provide a way to model "real-world" problems.[2]

[1] Even larger natural numbers for old ladies who live in shoes.

[2] Many of the problems discussed in this book are admittedly contrived and clearly fictional, but hopefully you will see how these toy problems can be generalized to actually represent problems that people would care about in reality.

To get a feel for the subject, let's consider the types of problems you solve in discrete math. Here are a few simple examples:

Investigate!

Note: Throughout the book you will see Investigate! *activities like this one. Answer the questions in these as best you can to give yourself a feel for what is coming next.*

1. The most popular mathematician in the world is throwing a party for all of his friends. To kick things off, they decide that everyone should shake hands. Assuming all 10 people at the party each shake hands with every other person (but not themselves, obviously) exactly once, how many handshakes take place?

2. At the warm-up event for Oscar's All-Star Hot Dog Eating Contest, Al ate one hot dog. Bob then showed him up by eating three hot dogs. Not to be outdone, Carl ate five. This continued with each contestant eating two more hot dogs than the previous contestant. How many hot dogs did Zeno (the 26th and final contestant) eat? How many hot dogs were eaten in total?

3. After excavating for weeks, you finally arrive at the burial chamber. The room is empty except for two large chests. On each is carved a message (strangely in English):

   Exactly one of these chests contains a treasure, while the other is filled with deadly immortal scorpions.

   For either chest, if the chest's message is true, then the chest contains treasure.

   The problem is, you don't know whether the messages are true or false. What do you do?

4. Back in the days of yore, five small towns decided they wanted to build roads directly connecting each pair of towns. While the towns had plenty of money to build roads as long and as winding as they wished, it was very important that the roads not intersect with each other (as stop signs had not yet been invented). Also, tunnels and bridges were not allowed, for moral reasons. Is it possible for each of these towns to build a road to each of the four other towns without creating any intersections?

As you consider the problems above, don't worry if it is not obvious to you what the solutions are. We are more interested here in what sort of information we need to be able to answer the questions. How can we represent the situation using individually separate and distinct objects? Don't read on until you have thought about at least this for each of the questions.

Ready? Here are some things you might have thought about:

1. The people at the party are individuals. We can consider the *set* of people. We can also consider sets of pairs of people, since it takes exactly two people to shake hands. So the question is really, how many pairs can you make using elements from a 10-element set?

   For example, if there were three people at the party, conveniently named 1, 2, and 3, then the pairs would be $(1, 2)$, $(1, 3)$, and $(2, 3)$. Or should we include $(2, 1)$, $(3, 1)$, and $(3, 2)$ as well?

2. To count the number of hot dogs eaten, either by an individual or in total, we could use a **sequence** of integers (whole numbers). The nth term in the sequence might represent the number of hot dogs eaten by the nth contestant. We can consider a second sequence, also of integers, that gives the total number of hot dogs eaten by the first n contestants combined.

   The solution to the problem will then be the value of the 26th term in the sequence. To help us find this, we could consider the rate of growth of the sequences, as well as how these two sequences relate to each other.

3. Logic questions also belong under the discrete math umbrella: Each statement can have a *value* of True or False (and there is nothing in-between). To answer questions like that of the chests of scorpions, we must understand the structure of the statements, and how the truth values of the parts of the statements interact to determine the truth value of the whole statement.

4. The last question is about a discrete structure called a **graph**, not to be confused with a graph of a function or set of points. We can use a graph to represent which elements of a set (or towns) are related to each other (or connected by a road). In this case, the question becomes, can we draw a graph with five vertices (towns) and ten edges (roads) such that no two edges intersect?

The four problems above illustrate the four main topics of this book: **combinatorics** (the theory of ways things *combine*; in particular, how to count these ways), **sequences**, **symbolic logic**, and **graph theory**. However, there are other topics that are also considered part of discrete mathematics, including computer science, abstract algebra, number theory, game theory, probability, and geometry (some of these, particularly the last two, have both discrete and non-discrete variants).

Ultimately the best way to learn what discrete math is about is to *do* it. Let's get started! Before we can begin answering more complicated (and fun) problems, we will consider a very brief overview of the types of discrete structures we will be using.

Reading Questions

Each section of the book will end with a small number of *Reading Questions* like the ones below. These are designed to help you reflect on what you have read. In particular, the final reading question asks you to ask a question of your own. Thinking about what you don't yet know is a wonderful way to further your understanding of what you do.

1. Right now, how would you describe what **discrete** mathematics is about, if you were telling your friends about the class you are in? Write one or two sentences.

2. What questions do you have after reading this section? Write at least one question about the content of this section that you are curious about.

Chapter 1

Expressions and Equations

Text before the first section.

1.1 Operations

Objectives

In this section, we learn how to:

1. Simplify an expression using the Order of Operations.
2. Analyze a non-constant expression using the Order of Operations.
3. Describe expressions and their parts using the words sum, term, product, and factor.
4. Construct geometric representations of addition and multiplication.
5. Justify major algebraic formulas geometrically.
6. Connect collecting like terms to the Distributive Law.

Algebra provides a powerful way of solving many kinds of questions Algebra achieves this by enhancing arithmetic. Arithmetic focuses on the result of operations like addition and multiplication when using numbers that are constant. The contribution of algebra is to enhance arithmetic with variables and with geometry, with a focus on how the numbers appearing a problem *relate* to each other. With these extra ways of thinking, algebra allows us to answer many questions that are really hard to think about using arithmetic alone.

I want your experience in math to be as smooth and as enjoyable as possible. I want this for you, even if you can recall being frustrated with mathematics. To achieve that goal, I am deliberately writing this book for you the student, and not for your teacher. I hope to take each part of algebra that has the potential to be challenging and *really* break it down step-by-step. This does mean that I may ask you to try something different from the way you have done it in the past. I might also ask you to think about things that you haven't really thought about much before. I hope you'll give it a shot: what do you have to lose by trying this subject in a new way? In fact, I encourage you to really think about the language used in mathematics.

Note 1.1.1 Pay attention to the language of mathematics.

As an example of this, when writing 2^x be sure to say one of the following:

- "2 to the x"
- "2 to the xth power"
- "2 raised to the x"
- "2 raised to the xth power"

If we only say "2" then slightly pause to say "x" this focuses on the specific individual symbols. The bigger problem is that saying "2" followed by "x" is taken to mean $2x$. Why is this a problem? Because 2^x and $2x$ are not equal. In fact, when $x = 3$ then 2^x simplifies to 8, but $2x$ simplifies to 6.

1.1.1 Order of Operations

We will take a little time to make sure that everyone is on the same page regarding the Order of Operations. Before we go any further, we should describe what we mean by an expression.

Definition 1.1.2

An **expression** is mathematical notation representing a number.

By the way... An expression may consist of just a single constant such as 3 or $\frac{4}{7}$, or can be a variable representing a number such as x or y, or can be a combination of constants and variables connected by operations such as addition, subtraction, multiplication, division, and exponentiation.

Example 1.1.3

Both -40 and $\sqrt{23}$ are examples of expressions. Both of these expressions are *constants*.

Example 1.1.4

Writing $-40 + \sqrt{64}$ is another expressions. This is an expression even if we haven't simplified this to the value -32. Like the two expressions in the previous example, the expression shown here (both the unsimplified and the simplified versions) is a *constant*, since the expression lacks any variable(s).

Example 1.1.5

Both $9x - 8$ and $x^2 + 3x + 31$ are examples of expressions which mention the variable x. In the first expression, the variable x is written once. In the second expression, the variable x is written twice.

Example 1.1.6

The expression $x^2 + 2x + y^2 - 6y$ mentions two variables. Since one or more variables appear, this expression is *not* a constant.

Please note that an expression does not contain an equal sign. For example, $x^2 + 2x + y^2 - 6y = 22$ is not an expression. The notation that we just wrote instead states that one exrpession is equal to another expression.

We need the Order of Operations because this it is easy to misinterpret an expression if we do not all agree on how to read expressions. The Order of Operations is a set of rules that tells us the order in which to evaluate (and more generally read) an expression.

Order of Operations.

An expression must always be simplified and read by following the **Order of Operations**:

1. Parentheses
2. Exponents
3. Multiplication and Division (from left to right)
4. Addition and Subtraction (from left to right)

In addition, every time you write an expression, ensure your writing is based on the Order of Operations.

Remark 1.1.7 By the time we get to the third part of the Order of Operations, all exponents would have been evaluated. At this point, we look for any multiplication or division. If there is both multiplication and division, we evaluate them from left to right. It is not true that multiplication must be done before division. All divisions and multiplications that we see have the same level of precedence, and we evaluate them scanning from left to right in that order.

Remark 1.1.8 Similarly, by the time we get to the last part of Order of Operations, all multiplications and divisions would have been handled. That means that what remains of our expressions should only have addition and subtraction operations remaining. These should be handled from left to right.

Remark 1.1.9 The Order of Operations is sometimes remembered by the acronym PEMDAS, which stands for Parentheses, Exponents, Multiplication, Division, Addition, and Subtraction.

Because of the way that PEMDAS is often taught, many people mistakenly believe that multiplication must be done before division, and addition must be done before subtraction. This is not true. For this reason, some people prefer the acronym GEMA, which stands for Grouping symbols, Exponents, Multiplication and Division, Addition and Subtraction. In the acronym GEMA, the "G" is used to indicate that there are many kinds of grouping symbols, not just parentheses. Also, the multiplication and division are addressed together in the "M", and addition and subtraction are addressed together in the "A".

Example 1.1.10

Simplify the expression $5^2 - 30 \div 3 + 2 \cdot 6$.

Solution. We will simplify the expression by following the Order of Operations. First, notice that there are no parentheses, so we move on to the next part of the Order of Operations. There is a place where the expression has exponens, so we zoom in on 5^2 and simplify this portion of the expression to 25. So, the expression given to us becomes $25 - 30 \div 3 + 2 \cdot 6$.

Now there is no more exponents. Next, we look for any multiplication or division. We see both multiplication and division, so we evaluate them from left to right. The left-most multplication or division we see is the division $30 \div 3$, which simplifies to 10. This gives us the expression $25 - 10 + 2 \cdot 6$. Continuing to scan from left to right for any multiplications or divisions, we see the multiplication $2 \cdot 6$, which simplifies to 12. This gives us the expression $25 - 10 + 12$.

Now there are no more multiplications or divisions, we look for any additions or subtractions, starting from the left. The left-most addition or subtraction we see is the subtraction $25 - 10$, which simplifies to 15. This gives us the expression $15 + 12$. Continuing to scan from left to right for any additions or subtractions, we see the addition $15 + 12$, which simplifies to 27.

Because this was our first example of applying the Order of Operations, we wanted to be very thorough to explain each step. To present our work, we start from the original expression and after writing an equal sign (to indicate that what we will write next is equal) write a simplified version of the expression. We continue this process until we reach the final simplified expression. For this example, we have $5^2 - 30 \div 3 + 2 \cdot 6 = 25 - 30 \div 3 + 2 \cdot 6 = 25 - 10 + 2 \cdot 6 = 25 - 10 + 12 = 15 + 12 = 27$.

It is also acceptable to write each expressions on their own lines, as follows:

$$\begin{aligned} 5^2 - 30 \div 3 + 2 \cdot 6 &= 25 - 30 \div 3 + 2 \cdot 6 \\ &= 25 - 10 + 2 \cdot 6 \\ &= 25 - 10 + 12 \\ &= 15 + 12 \\ &= 27 \end{aligned}$$

Note that when presenting our work *vertically* we still include the equal signs to indicate that each expression is equal to the previous expression.

Expectation.

When simplifying any expression, it is important to include equal signs to indicate that each expression is equal to the previous expression.

Expectation.

While simplifying expressions, ensure that the next expression you write is truly equal to the previous expression, instead of just writing the portion of the expression that is changing.

Example 1.1.11

Simplify the expression $3 \cdot (6 - 4)^2 \div 2$.

Solution 1. We can present our work horizontally, continuing to always write to the right of an equal sign like this $3 \cdot (6 - 4)^2 \div 2 = 3 \cdot 2^2 \div 2 = 3 \cdot 4 \div 2 = 12 \div 2 = 6$.

Solution 2. We can instead present our work vertically

$$\begin{aligned} 3 \cdot (6 - 4)^2 \div 2 &= 3 \cdot 2^2 \div 2 \\ &= 3 \cdot 4 \div 2 \\ &= 12 \div 2 \\ &= 6 \end{aligned}$$

Example 1.1.12

Simplify the expression $200 - 4^2 \div 8 \times 5 + 6$.

Solution 1. We can present our work horizontally, always writing to the right of the equal sign like this: $200 - 4^2 \div 8 \times 5 + 6 = 200 - 16 \div 8 \times 5 + 6 = 200 - 2 \times 5 + 6 = 200 - 10 + 6 = 190 + 6 = 196$.

Solution 2. We can instead present our work vertically:

$$\begin{aligned} 200 - 4^2 \div 8 \times 5 + 6 &= 200 - 16 \div 8 \times 5 + 6 \\ &= 200 - 2 \times 5 + 6 \\ &= 200 - 10 + 6 \\ &= 190 + 6 \\ &= 196 \end{aligned}$$

Habit.

Always read expressions based on the Order of Operations.

Habit.

Always write expressions based on the Order of Operations.

1.1.2 Order of Operations with variables

It is important for us to apply the Order of Operations not *only* to simplify expressions that contain only constants, but to also apply the Order of Operations when to interpret expressions that contain variables. This is important because one of the main contributions of algebra over arithmetic is to use variables to represent numbers that we do not know yet. So even when expressions contain variables, we still must read and write based on the Order of Operations. Doing this might feel new and totally weird, but let's explain how it's done and walk through examples together.

How to apply Order of Operations to expressions with variables.

Given an expression:

1. Identify each operation that is written in the given expression.
2. Apply the Order of Operations to identify in which order the operations *would* be performed.

Instead of simplifying an expression, we would hypothetically perform the operations.

Example 1.1.13

What order are the operations performed in the expression $z^3 - xy$?

Solution. The expression $z^3 - xy$ has three operations. Scanning from left to right, we see the following operations present: exponentiation, subtraction, and multiplication. When there is no symbol written between two variables, there is a hidden multiplication sign. In fact, to make it clearer, we can rewrite the expression as $z^3 - x \cdot y$.

Now that we have identified which operations are present in this expression (in this example, three of them) we will apply the Order of Operations to determine in which order these operations would be performed.

- Since there are no parentheses in our expression, the first operation that we would perform is the exponentiation. That is, if we knew the value of z, then we would first simplify z^3.
- Next, we would perform the multiplication $x \cdot y$. In other words, if we knew the value of x and we knew the value of y, then our work in this second step would be to simplify, and we'd have xy, "whatever the value of that is"
- Finally, we would perform the subtraction: we would take whatever the value of z^3 and subtract from this whatever the vaule of $x \cdot y$ is to get $z^3 - (x \cdot y)$.

Notice that we never simplified the expression $z^3 - xy$ or any part of this expression. We couldn't, because we did not know the numerical value of any of the variables. It seems like what we're doing is lazy, but I want to spin this into something positive. I encourage you to think of it this way: since we don't have the numbers behind the variables, we *get* to be lazy!

To perform this kind of analysis, it is helpful to say phrases like "whatever the value of *that* is". We are discussing in which order we would *hypothetically* perform the operations, if we knew the values of the variables. It seems like what we're doing is a bit silly, but it is important to practice this way of thinking. It is rare that I'll encourage the following, but I encourage you to say out loud the full text of the next several examples, pausing right before any phrase that looks similar to "whatever the value of *that* is" when this kind of phrase appears right after an expression followed by a comma.

Example 1.1.14

What order are the operations performed in the expression $(a + b) \cdot c^6$?

Solution. First, we identify the operations present, which are an addition, a multiplication written as a dot, and exponentiation. In addition, a part of the expression is contained inside parentheses.

- Since $a + b$ is in parentheses, we would perform the addition first. So, if we hypothetically knew the values of a and b, we'd replace this with the value of $a + b$, whatever value that is.
- After this, there would be no parentheses. Since all that would be left is a multiplication and an exponentiation, we would perform the exponentiation. We would take whatever the value of c is and raise this value to the 6th power. This would give us c^6, whatever the value of *that* is.
- Finally, we'd take whatever the value of $(a + b)$ is and whatever the value of c^6 is and multiply these values together.

Notice that we never simplified the expression $(a + b) \cdot c^6$ or any part of this expression. This process may feel weird to you because you may never have been asked about analyzing expressions in this way. Digging into the careful details of how to analyze an expression like this often gets overlooked but this next-level type of problem in applynig the Order of Operations sets up an important foundation for reading and writing expressions in algebra correctly. It's weird because it's new, but it's important because algebra enhances arithmetic by having variables, so I want you to be comfortable with reading and writing expressions with variables.

Example 1.1.15

What order are the operations performed in the expression $a(b - c)^4 + 7d$?

Solution. To make it clearer where the operations are, let's rewrite the expression as $a \cdot (b - c)^4 + 7 \cdot d$.

- Since $b - c$ is in parentheses, we would perform this subtraction first.
- Next, there would be no more parentheses, so we would perform the exponentiation: we would take $(b - c)$, whatever the value of $(b - c)$ is, and raise this to the 4th power. Thus, we'd have whatever the value of $(b - c)^4$ is.
- Next, we would perform the multiplication $a \cdot (b - c)^4$. That is, if we knew the value of a we would multiply this by the value of value of $(b - c)^4$ from the previous step, which would give us $a \cdot (b - c)^4$, "whatever the value of that is".
- Then, we would perform the multiplication $7 \cdot d$. That is, if we knew the value of d, we would simplify $7 \cdot d$.

- Finally, we would perform the addition: we would take whatever the value of $a \cdot (b-c)^4$ is and add to this whatever the value of $7 \cdot d$ is.

The activity we just went through leads up to the following new type of activity. We will be given two expressions which will only differ from each other in the inclusion or removal of a set of parentheses. Then, we will determine if the two expressions are equal, or if we don't know based only on the Order of Operations.

Example 1.1.16

Based only on the Order of Operations, are $x-(y \cdot z)$ and $x-y \cdot z$ equal or do we not have enough information?

Solution.

- In the first expression $x-(y \cdot z)$, we perform the multiplication $y \cdot z$ first because the multiplication is in parentheses. Then we perform the subtraction of x and $y \cdot z$ to get $x-(y \cdot z)$.

- In the second expression $x-y \cdot z$, there are no parentheses. We perform the multiplication $y \cdot z$ first. Then we perform the subtraction of x and $y \cdot z$ to get $x-y \cdot z$.

In both expressions, we perform the multiplication $y \cdot z$ first, and then we perform the subtraction of x and $y \cdot z$ next. Based only on the Order of Operations, the two expressions are equal.

Example 1.1.17

Based only on the Order of Operations, are $(a+b) \cdot c^6$ and $a+b \cdot c^6$ equal or do we not have enough information?

Solution.

- In the first expression $(a+b) \cdot c^6$, we perform the addition $a+b$ first because the the content is in parentheses. We do exponentiation next to get c^6. Our final operation is the multiplication of the value of $a+b$ and the value of c^6.

- In the second expression $a+b \cdot c^6$, we perform the exponentiation first to get the value of c^6. Next, we perform the multiplication of b and c^6. Finally, we perform the addition of a and $b \cdot c^6$.

In the first expression, we perform the addition first, then the exponentiation, and finally the multiplication. In the second expression, we perform the exponentiation first, then the multiplication, and finally the addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal. (These expressions might be equal or they might not, but we cannot determine this based only on the Order of Operations. If these expressions happened to be equal, this would have to be explained by something *other* than the Order of Operations.)

Example 1.1.18

Based only on the Order of Operations, are $a + b \cdot c - d$ and $a + (b \cdot c) - d$ equal or do we not have enough information?

Solution.

- In the first expression $a + b \cdot c - d$, we perform the multiplication $b \cdot c$ first. Then, we perform the addition of a and the value of $b \cdot c$. Finally, we perform the subtraction of the value of $a + b \cdot c$ and d.
- In the second expression $a + (b \cdot c) - d$, we perform the multiplication $b \cdot c$ first because this is in parentheses. Then, we perform the addition of a and the value of $b \cdot c$. Finally, we perform the subtraction of the value of $a + b \cdot c$ and d.

In both expressions, we perform the multiplication first, then the addition, and finally the subtraction. Based only on the Order of Operations, the two expressions are equal.

Example 1.1.19

Based only on the Order of Operations, are $a \div (b + c) \times 7$ and $a \div b + c \times 7$ equal or do we not have enough information?

Solution.

- In the first expression $a \div (b + c) \times 7$, we perform the addition $b + c$ first because the addition is in parentheses. Next, we perform the division of a by the value of $b + c$. Finally, we perform the multiplication of the value of $a \div (b + c)$ and 7.
- In the second expression $a \div b + c \times 7$, we perform the division $a \div b$ first. Next, we perform the multiplication $c \times 7$. Finally, we perform the addition of the value of $a \div b$ and the value of $c \times 7$.

In the first expression, we perform the addition first, then the division, and finally the multiplication. In the second expression, we perform the division first, then the multiplication, and finally the addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

Example 1.1.20

Based only on the Order of Operations, are $(w+x) \div (y+z)$ and $w+x \div y+z$ equal or do we not have enough information?

Solution.

- In the first expression $(w+x) \div (y+z)$, we perform the addition $w+x$ first because this is in parentheses. Next, we perform the addition $y+z$ because this is in parentheses. Finally, we perform the division of the value of $w+x$ by the value of $y+z$.

- In the second expression $w+x \div y+z$, we perform the division $x \div y$ first. Next, we perform the addition of w and the value of $x \div y$. Finally, we perform the addition of the value of $w+x \div y$ and z.

In the first expression, we perform the addition $w+x$ first, then the addition $y+z$, and finally do the division of whatever the value of $w+x$ is by whatever the value of $y+z$ is. In the second expression, we perform the division first, then an addition, and finally another addition. To be clear, in the first expression the value of $w+x$ is divided by whatever the value of $y+z$ is, and in the second expression the value of x is divided by whatever the value of y is. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

Example 1.1.21

Based only on the Order of Operations, are $a \cdot (b-c)^4 + 7 \cdot d$ and $a \cdot b - c^4 + 7 \cdot d$ equal or do we not have enough information?

Solution.

- In the first expression $a \cdot (b-c)^4 + 7 \cdot d$, we perform the subtraction $b-c$ first because the subtraction is in parentheses. Next, we perform the exponentiation to get $(b-c)^4$. Then, we perform the multiplication of a and $(b-c)^4$. Next, we perform the multiplication of 7 and d to get $7 \cdot d$. Finally, we perform the addition of the value of $a(b-c)^4$ and the value of $7 \cdot d$.

- In the second expression $a \cdot b - c^4 + 7 \cdot d$, we perform the exponentiation first to get the value of c^4. Next, we perform the multiplication of a and b. Then, we perform the multiplication of 7 and d. Finally, we perform the subtraction of the value of $a \cdot b$ and the value of c^4, and then add to this whatever the value of $7 \cdot d$ is.

In the first expression, we perform the subtraction first, then the exponentiation, then two multiplications, and finally an addition. In the second expression, we perform an exponentiation first, then two multiplications, then a subtraction, and finally an addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

1.1.3 Language

Before we move on, let's be sure that we're on the same page regarding some language used to describe expressions.

Definition 1.1.22

An expression that is the result of adding and/or subtracting two or more expressions together is called a **sum**. Each individual piece of the sum is called a **term**.

Informally, a *sum* is an expression that has additions and/or subtractions as its last operations. Consider using "mental scissors" to cut at each plus sign and at each minus sign. Then each of the pieces that we have is called a *term*.

Example 1.1.23

The expression $3x + 5y - 7$ is the result of adding and subtracting expressions together, so is called a sum. The three terms in this sum are $3x$, $5y$, and 7.

Definition 1.1.24

An expression that is the result of multiplying two or more expressions together is called a **product**. Each individual piece of the product is called a **factor**.

Informally, a *product* is an expression that has multiplications as its last operations. Consider using "mental scissors" to cut at each multiplication sign. (Before doing this, you may wish to draw in any hidden multiplication signs.) Then each of the pieces that we have is called a *factor*.

Example 1.1.25

The expression $4(x + y)z$ is the result of multiplying expressions together, so is called a product. The three factors in this product are 4, $x + y$, and z.

Example 1.1.26

Is $2a(b+c)(d+e)$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $2a(b+c)(d+e)$ is a product. The three factors in this product are $2a$, $(b+c)$, and $(d+e)$.

Example 1.1.27

Is $4x^2+3y-7\div z$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $4x^2+3y-7\div z$ is a sum. The three terms in this sum are $4x^2$, $3y$, and $7\div z$.

Example 1.1.28

Is $(a-b)^2+(c+d)^2$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $(a-b)^2+(c+d)^2$ is a sum. The two terms in this sum are $(a-b)^2$ and $(c+d)^2$.

1.1.4 Pictures

In algebra, we will often use pictures to represent expressions. This is a powerful way to think about expressions, and it is important to be able to translate between pictures and expressions. This may be new to you, but I encourage you to give it a try! Practicing this now will make future concepts go smoother!

Representing Addition Geometrically.

Suppose a and b are positive real numbers. Then $a+b$ is geometrically represented by the length of the stick made by gluing a stick of length a to a stick of length b .

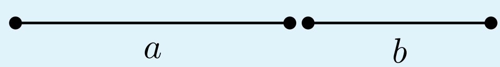

Figure 1.1.29 A picture of $a+b$

We have drawn the sticks slightly separated so that we can see them individually, but in reality we should imagine them pushed together. In the drawing, I made the choice to represent a as a slightly larger number than b. In addition, I chose to draw the stick representing a on the left and the stick representing b on the right because in the expression $a+b$, the a appears to the left of the plus sign, while the b appears to the right of the plus sign.

Example 1.1.30

Represent $3+4$ geometrically.

Solution.

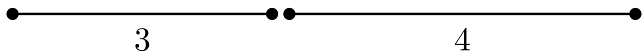

Figure 1.1.31 A picture of $3+4$

Example 1.1.32

Represent $2+0.5$ geometrically.

Solution.

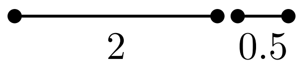

Figure 1.1.33 A picture of $2+0.5$

Example 1.1.34

Represent $x+2$ geometrically.

Solution.

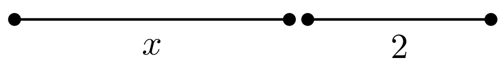

Figure 1.1.35 A picture of $x+2$

We can apply the geometric representation of addition to learn several important algebra facts about addition.

Commutative Property of Addition.

$$a+b=b+a$$

Example 1.1.36

Give a geometric explanation of why $a+b=b+a$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

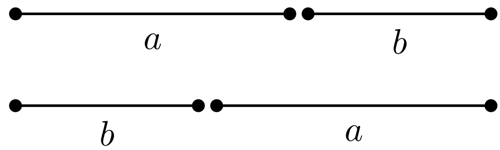

Figure 1.1.37 A picture of $a + b$ and $b + a$

The left side, $a + b$, is represented by a stick of length a glued to a stick of length b. The right side, $b + a$, is represented by a stick of length b glued to a stick of length a. In both cases, the resulting stick has the same length, so the two expressions are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b.)

In the Commutative Property of Addition $a + b = b + a$, we can substitute any expression we want for the a and any expression we want for the b. If we can for a moment explain what the Commutative Property of Addition is saying informally, the result of "this" plus "that" is the same as "that" plus "this". In other words, it is saying that when we add two things together, the order in which we add them doesn't matter. For example the Commutative Property of Addition tells us that $x^7 + 31y$ is equal to $31y + x^7$.

Associative Property of Addition.

$$(a + b) + c = a + (b + c)$$

Example 1.1.38

Give a geometric explanation of why $(a + b) + c = a + (b + c)$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

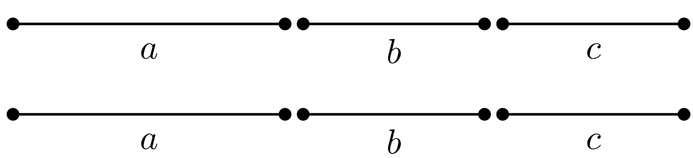

Figure 1.1.39 A picture of $(a + b) + c$ and $a + (b + c)$

The left side, $(a + b) + c$, is represented by a stick of length a glued to a stick of length b, and then this resulting stick is glued to a stick of length c. The right side, $a + (b + c)$, is represented by a stick of length b glued to a stick of length c, and then this resulting stick is glued to a stick of length a. (What the drawing doesn't indicate is the order of the gluing, so we need to clarify this with words: in the first diagram the stick of length a is glued to the stick of length b first, but in the second diagram the stick of length b is glued to the stick of length c first. If doing a hand drawing arrows can be drawn with labels like "glue here first" and "glue here next".) In both cases, the resulting stick has the same length, so the two expressions are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b or c.)

Having the Commutative Property of Addition and the Associative Property of Addition together basically tells us that when the expression we have is a sum, we can rearrange the terms in any order we want and simplify the addition of any terms in any order we want. For example $5a + 3b - 7c + 12d^8$ is equal to $12d^8 - 7c + 5a + 3b$ and is also equal to $-7c + 3b + 12d^8 + 5a$. When rearranging, be sure that any minus sign that was in front of a term stays in front of that term.

Representing Multiplication Geometrically.

Suppose a and b are positive real numbers. Then $a \cdot b$ is geometrically represented by the area of a rectangle with height a and width b .

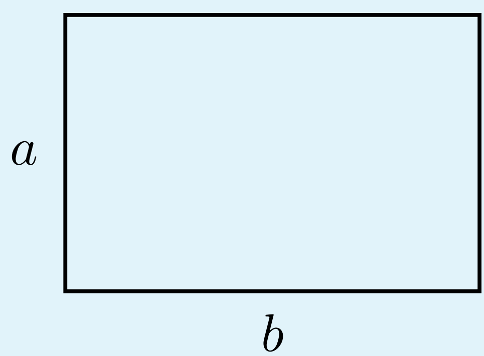

Figure 1.1.40 The area of the rectangle is a picture of $a \cdot b$

We will draw with the convention that the factor before the muliplication symbol is the height of the rectangle and the factor after the multiplication symbol is the width. In the drawing, I made the choice to represent a as a slightly smaller number than b.

Note 1.1.41 What we're introducing regarding the geometric representation of multiplication is an idea that we've seen before in geometry: we often write $A = \ell w$ to represent the area A of a rectangle with length ℓ and width w. Here, we're just using a and b instead of ℓ and w as the two factors.

Example 1.1.42

Represent $3 \cdot 4$ geometrically.

Solution.

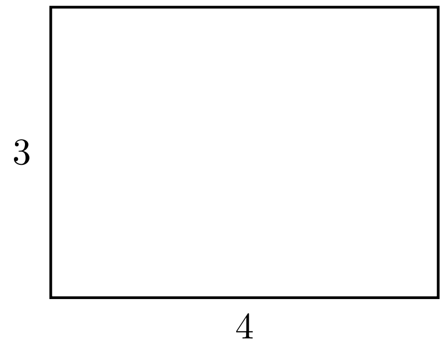

Figure 1.1.43 A picture of $3 \cdot 4$

Example 1.1.44

Represent $2 \cdot 0.5$ geometrically.

Solution.

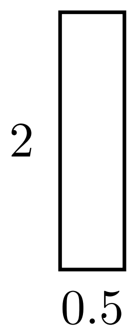

Figure 1.1.45 A picture of $2 \cdot 0.5$

We can apply the geometric representation of addition to learn several important algebra facts about addition.

Commutative Property of Multiplication.

$ab = ba$

Example 1.1.46

Give a geometric explanation of why $ab = ba$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

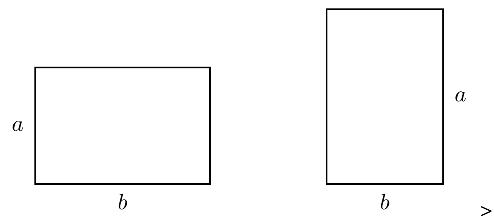

Figure 1.1.47 A picture of ab and ba

The left side, ab, is represented by a rectangle with height a and width b. The right side, ba, is represented by a rectangle with height b and width a. Both rectangles have the same area, since we can get from one rectangle to the other by rotation. Since the two areas are equal, the two expressions they represent, namely ab and ba, are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b.)

Associative Property of Multiplication.

$(ab)c = a(bc)$

We will skip the geometric explanation of the Associative Property of Multiplication, since we'd have to enhance our geometric interpretation of multiplication to include three factors. If you're curious, try to think through how you might represent $(ab)c$ and $a(bc)$ geometrically. As a hint, you'll need three-dimensional objects to do this.

Note 1.1.48 I just want to take a moment to keep encouraging you to think about addition and multiplication geometrically. It may be new and strange to you. It may seem like a waste of time to you. So far, it may just seem like a silly to give a geometric explanation of why certain facts (that might even feel obvious to you) are explained using geometry. However, setting up this foundation will make several challenging concepts in the future become a lot easier to digest.

How do I prevent confusing the two geometric representations?

The geometric representation of addition is the gluing together of sticks. The geometric representation of multiplication is the area of a rectangle.

One technique to help recall which is which is to recall that a usual presentation of the area formula is $A = \ell w$. Because this formula multiplies together two quantities (named ℓ and w), we can remember that multiplication is represented by area. Since addition is not represented by area, it must be represented by the other geometric idea we've seen, which is gluing sticks together.

Here's another techinque! Pick two numbers where the sum of the two numbers and the product of the two numbers is different. What I mean is that we wouldn't want to pick 2 and 2, since both the sum and the product are 4.

1. If we pick 2 and 3, we can ask ourselves what geometric object has some aspect of having size 5 and what geometric object has some aspect of having size 6. The object with size 5 is a stick of length 5, which is the result of gluing together a stick of length 2 and a stick of length 3. The object with size 6 is a rectangle with area 6, which is the result of multiplying together 2 and 3.

2. If we pick 4 and 7, we can ask ourselves what geometric object has some aspect of having size 11 and what geometric object has some aspect of having size 28. The object with size 11 is a stick of length 11, which is the result of gluing together a stick of length 4 and a stick of length 7. The object with size 28 is a rectangle with area 28, which is the result of multiplying together 4 and 7.

You can pick any two numbers you want, as long as the sum and product are different

We are now about to talk about an important algebra property that mentions both addition and multiplication. When we look at the geometric representation, we will need both geometric representations: gluing together of sticks *and* area of rectangles will *both* appear.

Distributive Law, version 1.

$a(b+c) = ab + ac$

Example 1.1.49

Give a geometric explanation of why $a(b+c) = ab + ac$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

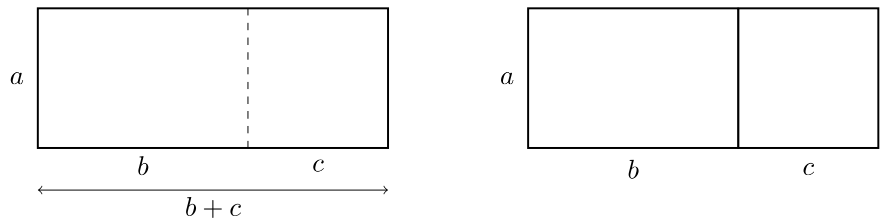

Figure 1.1.50 A picture of $a(b+c)$ and $ab+ac$

Before digging in, let's note that in the diagram the left, we see a stick of length b glued to a stick of length c, making a stick of length $b+c$. (This makes use of the geometric representation of addition.) The left side, $a(b+c)$, is represented by a rectangle with height a and width $b+c$. The right side, $ab+ac$, is represented by two rectangles: one with height a and width b, and the other with height a and width c. The area of the big rectangle on the left is equal to the sum of the areas of the two rectangles on the right, since we can get from the rectangle on the left to the two rectangles on the right by cutting the rectangle on the left vertically into two pieces. Since the area of the big rectangle on the left is equal to the sum of the areas of the two rectangles on the right, the two expressions they represent, namely $a(b+c)$ and $ab+ac$, are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b or c.)

In the Distributive Law $a(b+c) = ab+ac$, we can substitute any expression we want for the a, any expression we want for the b, and any expression we want for the c. If we can for a moment explain what the Distributive Law is saying informally, the result of "this" times the sum of "that" and "the other thing" is the same as "this" times "that" plus "this" times "the other thing". For example the Distributive Law tells us that $x(3y+5)$ is equal to $3xy+5x$.

Note 1.1.51 When you see $x(3y+5)$, I encourage you to think about the Distributive Law as we wrote it $a(b+c) = ab+ac$, and relate the a and b and c in the formula to the x and $3y$ and 5 in the expression $x(3y+5)$.

It is easy to dismiss this advice and just think "What's the point of all this? I can just see that I should distribute x." However, it is important to see exactly what this Distributive Law $a(b+c) = ab+ac$ is saying, and what it is *not* saying. The left side $a(b+c)$ addresses *only* an expression that has addition on the inside of the parentheses, and multiplication outside. For example, the formula $a(b+c) = ab+ac$ has *nothing* to say about the expression $j + (k \cdot \sqrt{m})$. It would be tempting to look at $j + (k \cdot \sqrt{m})$ and try to "distribute" the j somehow. But when we read the left side of $a(b+c) = ab+ac$ and we see that the left side says $a(b+c)$, we have to carefully note that addition is inside the parentheses with multiplication outside. The problem with $j + (k \cdot \sqrt{m})$ is that multiplication is inside the parentheses with the addition outside. So the Distributive Law $a(b+c) = ab + ac$ does not apply to $j + (k \cdot \sqrt{m})$.

Example 1.1.52

Use the Distributive Law to rewrite $5(x+2)$.

Solution. Using the Distributive Law $a(b+c) = ab + ac$, we can let $a = 5$, $b = x$, and $c = 2$. Then we have

$$5(x+2) = 5 \cdot x + 5 \cdot 2 = 5x + 10.$$

Example 1.1.53

Use the Distributive Law to rewrite $10(h^2 + t^3)$

Solution. Using the Distributive Law $a(b+c) = ab + ac$, we can let $a = 10$, $b = h^2$, and $c = t^3$. Then we have

$$10(h^2 + t^3) = 10 \cdot h^2 + 10 \cdot t^3 = 10h^2 + 10t^3.$$

In the example above, we informally say that we "**distributed** 10". We can also turn an expression of the form $ab + ac$ into an expression of the form $a(b+c)$. This process is called **factoring**. (Factoring is the reverse of the process of distributing.)

Example 1.1.54

Use the Distributive Law to factor $4c + 4d$.

Solution. Using the Distributive Law $a(b+c) = ab + ac$, we can let $a = 4$, $b = c$, and $c = d$. Then we have

$$4c + 4d = 4(c + d).$$

Example 1.1.55

Use the Distributive Law to factor $12x^2 + 18x$.

Solution. Using the Distributive Law $a(b+c) = ab + ac$, we can let $a = 6x$, $b = 2x$, and $c = 3$. Then we have

$$12x^2 + 18x = 6x \cdot 2x + 6x \cdot 3 = 6x(2x + 3).$$

The selection of $6x$ occured by looking for the greatest common factor of $12x^2$ and $18x$.

Example 1.1.56

Use the Distributive Law to factor $15xy + 6y$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 3y$, $b = 5x$, and $c = 2$. Then we have

$$15xy + 6y = 3y \cdot 5x + 3y \cdot 2 = 3y(5x + 2).$$

The selection of $3y$ occured by looking for the greatest common factor of $15xy$ and $6y$.

Example 1.1.57

Use the Distributive Law to factor $9x + 9$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 9$, $b = x$, and $c = 1$. Then we have

$$9x + 9 = 9(x + 1).$$

Warning 1.1.58 When factoring $9x + 9$ it is common to make the mistake of saying that this expression is equal to $9(x)$ instead of $9(x + 1)$. When we see $9x + 9$, to successfully factor out a 9 we will write an expression in the format $9(?+?)$, and the first question mark is filled in by asking "9 times *what* is the first term $9x$", while the second question mark is filled in by asking "9 times *what* is the second term 9".

To provide a little more convincing, you can always check your factoring by taking your answer and distributing. Note that distributing in the expression $9(x + 1)$ gives us $9x + 9$.

Example 1.1.59

Factor the expression $71x + 71$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 71$, $b = x$, and $c = 1$. Then we have

$$71x + 71 = 71(x + 1).$$

As we factor out 71, the reason that there is a 1 as the second term in parentheses is because this answers "71 times *what* equals the second term 71 in the original expression", which applies the same reasoning for why an x appears as the first term in parentheses because this answers "71 times *what* equals the first term $71x$ in the original expression".

Distributive Law, version 2.

$a(b - c) = ab - ac$

This version has subtraction inside the parentheses instead of addition inside the parentheses. Since we didn't provide a geometric representation of subtraction, we won't provide a geometric explanation of why this version of the Distributive Law is true. (However, for a mental challenge, we can use the first version of the Distributive Law to explain why this version is true. We can think of $b - c$ as $b + (-c)$, and then apply the first version of the Distributive Law.)

Example 1.1.60

Use the Distributive Law to rewrite $4(x - 3)$.

Solution. Using the Distributive Law $a(b - c) = ab - ac$, we can let $a = 4$, $b = x$, and $c = 3$. Then we have

$$4(x - 3) = 4 \cdot x - 4 \cdot 3 = 4x - 12.$$

Example 1.1.61

Use the Distributive Law to rewrite $30x - 42x^2$.

Solution. Using the Distributive Law $a(b - c) = ab - ac$, we can let $a = 6x$, $b = 5$, and $c = 7x$. Then we have

$$30x - 42x^2 = 6x \cdot 5 - 6x \cdot 7x = 6x(5 - 7x).$$

The selection of $6x$ occured by looking for the greatest common factor of $30x$ and $42x^2$.

Distributive Law, version 3.

$(b + c)a = ba + ca$

This version has the multiplication on the right instead of the left.

Try it 1.1.62

Spend a few minutes providing the geometric explanation of why $(b + c)a = ba + ca$ is true. When doing this, remember we shouldn't select specific numbers for a, b, or c.

Distributive Law, version 4.

$(b-c)a = ba - ca$

Example 1.1.63

Use the Distributive Law to rewrite $7 - 14y$.

Solution. Using the Distributive Law $(b-c)a = ba - ca$, we can let $a = 7$, $b = 1$, and $c = 2y$. Then we have

$$7 - 14y = 1 \cdot 7 - 2y \cdot 7 = (1 - 2y)7.$$

Example 1.1.64

Use the Distributive Law to rewrite $13x + 7x$.

Solution. Using the Distributive Law $(b+c)a = ba + ca$, we can let $a = x$, $b = 13$, and $c = 7$. Then we have

$$13x + 7x = (13 + 7)x = 20x.$$

Example 1.1.65

Use the Distributive Law to rewrite $13x - 7x$.

Solution. Using the Distributive Law $(b-c)a = ba - ca$, we can let $a = x$, $b = 13$, and $c = 7$. Then we have

$$13x - 7x = (13 - 7)x = 6x.$$

The last two examples show that the process known as **collecting like terms** is actually justified by the Distributive Law. In fact, the only reason why collecting like terms works at all is because of the Distributive Law! Collecting like terms is actually factoring out the variable from several terms (though we often skip writing that "middle step" such as $(13 - 7)x$ and just simplify the part that is in parentheses in our head.)

Note 1.1.66 It may be tempting to ignore this comment about the connection between collecting like terms and the Distributive Law. If we ignore the connection, it would be easy to see an expression like $3x + \sqrt{7}\,x$ and feel stuck thinking that we "can't" collect like terms. However, if we remember that collecting like terms is actually factoring, then we can see that

$$3x + \sqrt{7}\,x = (3 + \sqrt{7})x.$$

In another example like $3x + 5x = (3 + 5)x = 8x$ we very often skip writing the middle step $(3 + 5)x$ and just directly go from $3x + 5x$ to $8x$. However, it's good to remember that there is a middle step, and that middle step is justified by the Distributive Law. Recalling this allows us to not get intimidated by expressions like $3x + \sqrt{7}\,x$.

We have shown four versions of the Distributive Law. Here is an extended version of the Distributive Law:

Distributive Law, one extended version.

$$a(b + c + d) = ab + ac + ad$$

Try it 1.1.67

Come up with other extended versions of the Distributive Law. Be careful to pay attention to where the addition and subtraction signs are, and where the multiplication signs are (including hidden multiplication signs).

1.1.5 SUMMARY

Expressions with and without variables can be interpreted and analyzed using the Order of Operations, and expressions without variables can be simplified down to a single constant using the Order of Operations. The result of adding and subtracting expressions called terms results in a sum, while the result of multiplying expression called factors results in a product. The geometric representation of addition by gluing together sticks and the geometric representation of multiplication by the area of a rectangle can be used to justify major algebraic formulas, including the Distributive Law. It is the Distributive Law which actually justifies the shortcut process known as collecting like terms.

1.1.6 EXERCISES

1. Write in your own words how each expression could be spoken aloud with clarity.

   (a) $8 + a$

   (b) $8a$

   (c) 8^a

   (d) $5x + 6^x$

   (e) $3^x - 3x$

   (f) $5(x + y)^2$

   (g) $5x + y^2$

(h) $\dfrac{3}{a+b}$

(i) $\dfrac{3}{a}+b$

2. Simplify the following expressions. Because there are no variables, each expression can be simplified based only on the Order of Operations. Because the intention of this exercise is to practice the Order of Operations, avoid applying the Distributive Law, even if you are certain that the Distributive Law could be applied.

   (a) $4(2+6)$

   (b) $4(2\cdot 6)$

   (c) $4-(3+5\cdot 2)+1$

   (d) $6-2^3$

   (e) $1-2-3-4-5-6$

   (f) $(2\cdot 3)^3$

   (g) $100-2\cdot 3\times 4$

   (h) $5(4+3)^2$

   (i) $10\div 2+3\cdot 4$

   (j) $9-2\times 3+5\times 10$

   (k) $20-12\div 4+2\times 5$

   (l) $7^2-3(4+1)+10\cdot 9$

3. In each expression below, identify which operation is performed first, which is performed next, and so on.

   (a) $a(b+c)^2$

   (b) $p\div q+r\cdot s$

   (c) $a-b\div c+d\times e$

   (d) $a^b-c(d+e)+x\cdot y$

   (e) $a\div b-(c+d)^m$

   (f) $p\div q-r\cdot s+2^m$

4. In each expression below, identify which operation is performed first.

   (a) $5+2^x$

   (b) $8(x-1)$

   (c) $8x-1$

   (d) $9\cdot(x-2)+4$

   (e) $a-b\div c+(x-y)^z$

   (f) $a-b\div c+x-y^z$

(g) $x \div y - z + a^b$
(h) $x \div (y - z) + a^b$
(i) $a \cdot b + c \div d - e \cdot f + g \cdot h$

5. In each expression below, identify which operation is performed last.

(a) $5 + 2^x$
(b) $8(x - 1)$
(c) $8x - 1$
(d) $9 \cdot (x - 2) + 4$
(e) $a - b \div c + (x - y)^z$
(f) $a - b \div c + x - y^z$
(g) $x \div y - z + a^b$
(h) $x \div (y - z) + a^b$
(i) $a \cdot b + c \div d - e \cdot f + g \cdot h$

6. In each part below, you are given two expressions. Based only on the Order of Operations, are the two expressions equal or do we not have enough information? Notice how similar the expressions are: in most cases, one expression just has more parentheses. (Base your answer *only* on the Order of Operations, and not on whether anything "should" or "shouldn't" distribute.)

(a) $3x + 5$ and $3(x + 5)$
(b) $2 + (a \times 6)$ and $2 + a \times 6$
(c) $(a + b)(c + d)$ and $a + b(c + d)$
(d) $(p \times q) + (r \times s)$ and $p \times q + r \times s$
(e) $(x^2 + 3)(2) - (5x - 7)(4)$ and $(x^2 + 3)2 - (5x - 7)4$
(f) $(t^4 - 3) \cdot (8) + (10)(x - 5)$ and $t^4 - 3 \cdot (8) + 10(x - 5)$
(g) $9 \cdot (x - 2) + 4$ and $9 \cdot x - 2 + 4$
(h) $a - b \div c + (x - y)^z$ and $a - b \div c + x - y^z$
(i) $x \div y - z + a^b$ and $x \div (y - z) + a^b$
(j) $a \cdot b + c \div d - e \cdot f + g \cdot h$ and $a \cdot b + c \div d - (e \cdot f + g \cdot h)$
(k) $p \div q + r \cdot s$ and $(p \div q + r) \cdot s$

7. Each expression below is a sum. State the terms.

(a) State the terms of $3x + 5 + 71y$
(b) State the terms of $4 + 2m + 3x + r$
(c) State the terms of $7a^2 - 3b + 4c - 9$
(d) State the terms of $5x - 7y + 2z + 9 + w$

8. What term(s) do $3x + 5 + 71y$ and $2m + 3x + r - 8y$ have in common?

9. Geometrically represent the following expressions.

   (a) $x + 3$
   (b) $p \cdot q$
   (c) $a + b + 4$
   (d) $p(q + r)$
   (e) $(a + b)(c + d)$
   (f) $(x + 3)(x + 3)$

10. For each geometric representation below, write an expression it represents. (Note, there is often more than one correct answer.)

   (a) $x + y + z$
   (b) $r \times s$
   (c) $p(q + r)$
   (d) $(p + q)r$
   (e) $(p + q)(r + s)$
   (f) $a(b + c + d)$
   (g) $a \times a$
   (h) $(x + 5)(x + 5)$

11. Factor each expression below.

   (a) $6 + 3x$
   (b) $10a - 30$
   (c) $9xy + 4xy$
   (d) $10x + 3x$
   (e) $10x + \sqrt{3}x$
   (f) $7x + x$
   (g) $23x + 23$

12. Simplify each expression below two ways: once by applying the appropriate Distributive Law to factor (showing all steps), and once by quickly collecting like terms. Why ask you to simplify each expression the "slightly slower way" and the "slightly faster way"? The purpose of this exercise is to reinforce the important idea that collecting like terms is actually justified by the Distributive Law.

   (a) $4x + 5x$
   (b) $13y - 2y$
   (c) $3a + 4a + 5a$
   (d) $9c - 2c + 4c$
   (e) $7z - z$

1.2 Expressions with Fractions

Objectives

In this section, we learn how to:

1. Add, subtract, multiply, and divide fractions.
2. Determine when common denominators are required and when they are not.
3. Determine when cancelling is permissible in a fraction and when it isn't.
4. Convert between mixed fractions and improper fractions.

1.2.1 Applications

Operations on fractions are a little more involved than operations on whole numbers, but what we learn will enable us to answer these questions:

1. A baking recipe calls for $5\frac{1}{4}$ cups of flour. The recipe says that it serves 10. You are hosting a party where 35 people will attend, so you need to make $3\frac{1}{2}$ times the recipe. How many cups of flour do you need?
2. You have a two-by-four piece of lumber that measures $2\frac{5}{8}$ feet long and another that is $1\frac{3}{4}$ feet long. You will glue the two pieces together to create one leg of a nightstand. But you'll need to go to the store to make three other legs of the same height. How long is each leg?
3. A swimming pool is being filled at a rate of $8\frac{1}{4}$ gallons per hour. How many gallons of water will be in the pool after $2\frac{1}{4}$ hour?
4. A car is traveling at a speed of 60 miles per hour. How far will the car travel in $4\frac{1}{2}$ hours?
5. A group of 5 friends go out to dinner. The bill comes to $\$48.75$. If they split the bill evenly, how much does each person pay?
6. A recipe calls for $\frac{3}{4}$ cup of sugar. You only have a $\frac{1}{3}$ cup measuring cup. How many $\frac{1}{3}$ cups of sugar do you need to use to get the correct amount?
7. A recipe calls for $\frac{3}{4}$ cup of sugar. You put in $\frac{1}{3}$ cup of sugar by accident. How much more sugar do you need to put in to get back on track in following the recipe?

1.2.2 Representation and Equal/Unequal Fractions

Representing Fractions Geometrically.

Suppose a and b are positive real numbers. We can represent the fraction $\frac{a}{b}$ as follows: Start with a pizza that has b equal-sized slices. Then $\frac{a}{b}$ is represented by eating a of those slices.

In this way, 1 represents taking/eating the whole pizza, and $\frac{1}{2}$ represents taking a pizza that only has 2 slices (those are *huge* slices, since there are only two of them) and eating one of those slices.

Example 1.2.1

The fraction $\frac{2}{7}$ represents taking a pizza that has been cut into 7 equal slices and eating 2 of those slices.

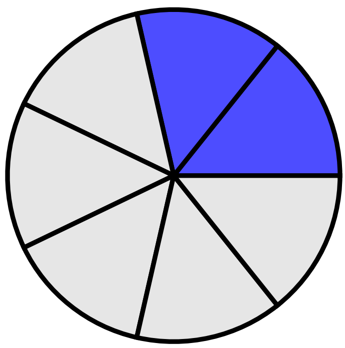

Figure 1.2.2 A picture of $\frac{2}{7}$

Example 1.2.3

What fraction is equal to $\frac{1}{3}$ but has 6 in the denominator? Draw pictures of $\frac{1}{3}$ and the resulting fraction to convince yourself.

Solution. The fraction $\frac{1}{3}$ represents taking a pizza that has been cut into 3 equal slices and eating 1 of those slices.

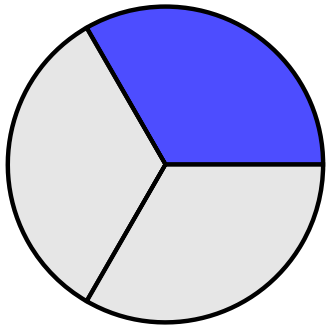

Figure 1.2.4 A picture of $\frac{1}{3}$

To get a denominator of 6, we need to cut each of the 3 slices in half. This gives us 6 equal slices, and we have eaten 2 of those slices. So the answer is $\frac{2}{6}$.

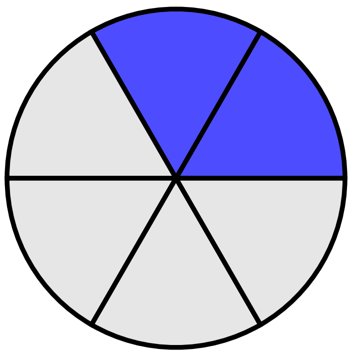

Figure 1.2.5 A picture of $\frac{2}{6}$

If we step back and ignore where the cuts were made by a pizza cutter, the blue shaded part (representing the eaten pizza) looks to be the same amount in both pictures. So $\frac{1}{3} = \frac{2}{6}$.

Example 1.2.6

What fraction is equal to $\frac{3}{4}$ but has 8 in the denominator? Draw pictures of $\frac{3}{4}$ and the resulting fraction to convince yourself.

Solution. The fraction $\frac{3}{4}$ represents taking a pizza that has been cut into 4 equal slices and eating 3 of those slices.

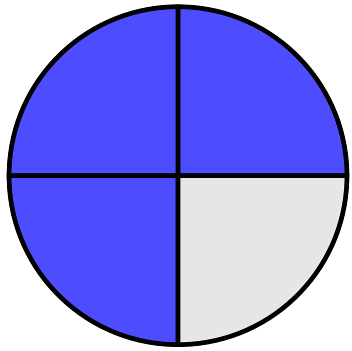

Figure 1.2.7 A picture of $\frac{3}{4}$

To get a denominator of 8, we need to cut each of the 4 slices in half. This gives us 8 equal slices, and we have eaten 6 of those slices. So the answer is $\frac{6}{8}$.

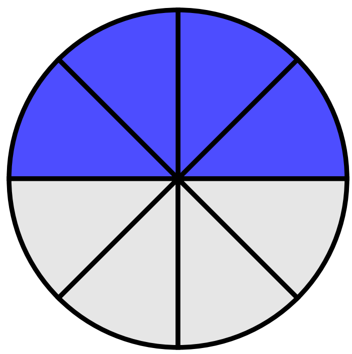

Figure 1.2.8 A picture of $\frac{6}{8}$

Try it 1.2.9

What fraction is equal to $\frac{2}{5}$ but has 15 in the denominator? Draw pictures of $\frac{2}{5}$ and the resulting fraction to convince yourself.

Note 1.2.10 In earlier examples, we had to cut each original slice of pizza into two equal-sized smaller pieces. In the *Try it* exercise, we need to cut each original slice of pizza into three equal-sized smaller pieces.

Note 1.2.11 In the *Try it* exercise, going from a total of 5 slices to a total of 15 slices meant that in the resulting pizza, every slice was very small. This makes intuitive sense to us: when the denominator is large (in other words, there are many slices), then the size of each slice is small. In addition, when the denominator is small (in other words, there are few slices), then the size of each slice is large.

To create an equal fraction to the fraction we're given but without drawing pictures, we can notice that we take our starting fraction and *multiply* both the numerator and denominator by the same number. To formally state this fact:

Equal Fractions, different denominators.

$$\frac{a}{b} = \frac{a \cdot c}{b \cdot c},$$

where b and c are nonzero.

Example 1.2.12

Using the Equal Fractions formula, What fraction is equal to $\frac{3}{5}$ but has 20 in the denominator?

Solution. To get a denominator of 20, we need to multiply the denominator of $\frac{3}{5}$ by 4. So we also need to multiply the numerator by 4.

$$\frac{3}{5} = \frac{3 \cdot 4}{5 \cdot 4} = \frac{12}{20}.$$

In what we just wrote, we went from the initial fraction to the middle expression to emphasize the Equal Fractions formula, though if we feel comfortable skipping this, we might write $\frac{3}{5} = \frac{12}{20}$ directly. That said, we may find a future example more complicated and writing the middle step (which *indicates* multiplication on top and on bottom, but does not *simplify* the multiplication) is good to practice, even in cases where we might not feel we need it.

Example 1.2.13

Using the Equal Fractions formula, What fraction is equal to $\frac{2}{3}$ but has 12 in the denominator?

Solution. To get a denominator of 12, we need to multiply the denominator of $\frac{2}{3}$ by 4. So we also need to multiply the numerator by 4.

$$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}.$$

Warning 1.2.14 When writing our work, we must be careful how we write our work. It is possible to write something that can be interpreted incorrectly. In detail:

- We can write
$$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}$$
exactly as the previous example showed.

- We can write the 4 that we were multiplying on top and on bottom before the original content like this:
$$\frac{2}{3} = \frac{4 \cdot 2}{4 \cdot 3} = \frac{8}{12}$$
and this just swaps the order of the factors in both multiplications.

- We can skip actually writing in the "times 4" on top and on bottom and just write in the resulting numerator and denominator like this:
$$\frac{2}{3} = \frac{8}{12}$$

- We *cannot* write
$$4\frac{2}{3}$$
to mean that we are multiplying by 4 on top and on bottom. There are two ways of interpreting $4\frac{2}{3}$. We will discuss one of the ways to interpret $4\frac{2}{3}$ by the end of this section, but a common way to interpret $4\frac{2}{3}$ is as a mixed fraction, whose value is clearly larger than 4 itself, while the original fraction $\frac{2}{3}$ is smaller than 1. So, we either jump to writing $\frac{8}{12}$ directly, or we need to write that we are multiplying by 4 on top and on bottom (and that really requires physically writing two 4s: one on top and one on bottom), but we cannot just write a single 4. Reading $4\frac{2}{3}$ as a mixed fraction has a differnt value than the original fraction, and in the other interpretation will also result in a value that is not equal to $\frac{2}{3}$.

For another example of this warning, if we start with $\frac{2}{7}$ and we want to properly write about multiplying by 10 on top and on bottom, we can write $\frac{2 \cdot 10}{7 \cdot 10}$ or $\frac{10 \cdot 2}{10 \cdot 7}$ or just immediately write $\frac{20}{70}$, but we cannot write $10\frac{2}{7}$.

Example 1.2.15

What fraction is equal to $\frac{2}{3}$ but has $3x$ in the denominator?

Solution. We need to multiply the denominator by x, so we also multiply the numerator by x.
$$\frac{2}{3} = \frac{2x}{3x}$$

Example 1.2.16

What fraction is equal to $\frac{2}{3}$ but has $3(10x+2)$ in the denominator?

Solution. We need to multiply the denominator by $(10x+2)$ so we also multiply the numerator by $(10x+2)$.

$$\frac{2}{3} = \frac{2(10x+2)}{3(10x+2)}$$

Here, if we wanted, we can distribute in both the numerator and denominator. However, we stopped where we did because we wanted to highlight the role of the Equal Fractions formula.

Example 1.2.17

What fraction is equal to $\frac{5x}{2y}$ but has $10y$ in the denominator?

Solution. We need to multiply the denominator by 5, so we also multiply the numerator by 5.

$$\frac{5x}{2y} = \frac{5 \cdot 5x}{5 \cdot 2y} = \frac{25x}{10y}$$

In certain situations, we can also use the Equal Fractions formula to reduce the size of the denominator.

Reducing Fractions.

If we see the same factor in the numerator and denominator of a fraction, the process of removing this common factor is called **cancelling** in a fraction or **reducing** a fraction. This is applying our earlier formula $\frac{a}{b} = \frac{a \cdot c}{b \cdot c}$ "backwards" in other words, this is turning $\frac{a \cdot c}{b \cdot c}$ into $\frac{a}{b}$

Example 1.2.18

What fraction is equal to $\frac{8}{20}$ but has 5 in the denominator?

Solution. The numerator and denominator both have a factor of 4. For the first several examples, we will intentionally slow down and rewrite the numerator and denominator in factored form to highlight the common factor, which helps highlight exactly the role taken by the *Reducing Fractions* process we just described.

$$\frac{8}{20} = \frac{2 \cdot 4}{5 \cdot 4} = \frac{2}{5}.$$

We cancelled the common factor of 4 in the numerator and denominator.

Example 1.2.19

What fraction is equal to $\frac{12}{27}$ but has 9 in the denominator?

Solution.

$$\frac{12}{27} = \frac{4 \cdot 3}{9 \cdot 3} = \frac{4}{9}.$$

We cancelled the common factor of 3 in the numerator and denominator.

Example 1.2.20

What fraction is equal to $\frac{18x}{24y}$ but has $4y$ in the denominator?

Solution.

$$\frac{18x}{24y} = \frac{3x \cdot 6}{4y \cdot 6} = \frac{3x}{4y}.$$

Now we can cancel the common factor of 6 in the numerator and denominator.

Example 1.2.21

Reduce the fraction $\frac{3x^2}{6xy}$ as much as possible.

Solution.

$$\frac{3x^2}{6xy} = \frac{x \cdot 3x}{2y \cdot 3x} = \frac{x}{2y}.$$

Now we can cancel the common factor of $3x$ in the numerator and denominator.

Warning 1.2.22 While we can cancel common *factors* in a fraction, we cannot cancel common *terms*. In other words,

$$\frac{a+c}{b+c} \neq \frac{a}{b}.$$

Let's describe this frequent error through two examples:

- If someone took $\frac{2+1}{4+1}$ and tried to "cancel" the $+1$ in the numerator and denominator, they would get $\frac{2}{4}$, which is 0.5 in a calculator. However, the original fraction can naturally be rewritten as $\frac{3}{5}$, which is 0.6 in a calculator.

- The issue is more likely to occur when there are variables. (In fact, the previous example with its decimal representations of fractions was included to create a concrete and convincing example that we cannot cancel terms on top and bottom.) We cannot turn $\frac{x+2}{x+3}$ into $\frac{2}{3}$ by "cancelling" the common x in the numerator and denominator. This is because x is a *term* on top and bottom, and not a *factor* on top and bottom.

Example 1.2.23

Reduce the fraction $\frac{20x+60}{14x+30}$

Solution 1. There is a temptation to want to cancel part of the 60 with the 30, but these are terms. We can only cancel factors. First, we rewrite the numerator and denominator in factored form:

$$\frac{20x+60}{14x+30} = \frac{2(10x+30)}{2(7x+15)} = \frac{10x+30}{7x+15}.$$

We cancelled the common factor of 2 in the numerator and denominator.

Solution 2. You may prefer to factor a larger factor out of the numerator. (Note, we still cannot cancel terms.)

$$\frac{20x+60}{14x+30} = \frac{20(x+3)}{2(7x+15)} = \frac{10(x+3)\cdot 2}{(7x+15)\cdot 2} = \frac{10(x+3)}{7x+15}.$$

After factoring, we replaced the 20 that we factored out with $10 \cdot 2$ to really highight that it is a factor of 2 on top and bottom that we cancelled. To make the final expression in this solution look like the final expression in the previous solution, we could distribute the 10 in the numerator. (In this example, you may feel comfortable skipping the writing the third expression, jumping directly from the second expression to the fourth expression. We included the third expression because going from the third expression to the fourth expression highlights the role of the *Reducing Fractions* process we described earlier.)

1.2.3 Adding and subtracting fractions

Try it 1.2.24

The pictures of $\frac{5}{6}$ and $\frac{4}{9}$ are shown below. How can we use these pictures to represent a picture that represents the value of $\frac{5}{6} + \frac{4}{9}$ as a single fraction?

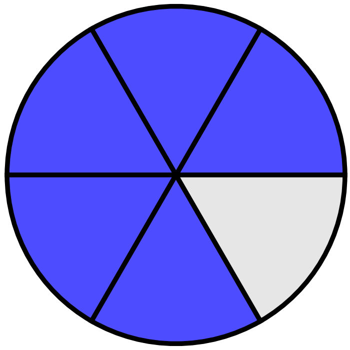

Figure 1.2.25 A picture of $\frac{5}{6}$

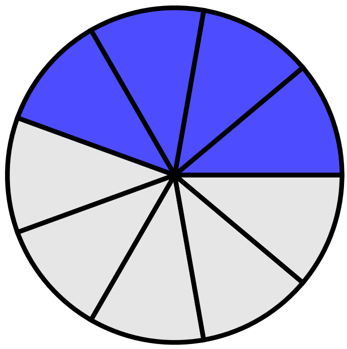

Figure 1.2.26 A picture of $\frac{4}{9}$

Did you try this exercise? What did you notice about the pictures? Because of the different denominators, the slices in the two pictures are different sizes, so it seems a little funny if we just tried to add the numerators across and add the denominators across when we write with notation. This hints at the following idea: To add fractions, we need a common denominator. In this case, we can use 18 as a common denominator. We can convert $\frac{5}{6}$ to a fraction with denominator 18 by multiplying the numerator and denominator by 3. We can convert $\frac{4}{9}$ to a fraction with denominator 18 by multiplying the numerator and denominator by 2. So:

$$\frac{5}{6} + \frac{4}{9} = \frac{5 \cdot 3}{6 \cdot 3} + \frac{4 \cdot 2}{9 \cdot 2} = \frac{15}{18} + \frac{8}{18} = \frac{15+8}{18} = \frac{23}{18}.$$

Note that the final answer is an improper fraction. This is perfectly fine. In algebra, it is often more helpful to leave answers as improper fractions. (By the end of this section, we'll explain exactly why improper fractions are preferred.)

Adding Fractions.

Simplifying the addition of fractions requires having a *common denominator*.

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}.$$

Notice in the two fractions on the left side of the formula above, c is the denominator for both fractions. The fact that c is written for both denominators on the left is the formula communicating to us that both fractions have to have the same denominator. When we have that common denominator c, the right side of this formula which says $\frac{a+b}{c}$ is tell us that fraction that we get as a result from simplifying the addition copies that same denominator c, while the numerator of the new fraction is made by adding the numerators of the two fractions that had the same denominator. It is incorrect to try to "just add straight across":

Warning 1.2.27

$$\frac{p}{q} + \frac{r}{s} \neq \frac{p+r}{q+s}$$

To show why this warning is here, in Try it 1.2.24, we saw $\frac{5}{6} + \frac{4}{9}$ simplifies to $\frac{23}{18}$ but if we added straight across (using the "fake formula" meantion in the warning), we would have gotten $\frac{9}{15}$. You can check that $\frac{23}{18}$ and $\frac{9}{15}$ are different answers with a calculator, or without a calculator, we can notice that $\frac{23}{18}$ is greater than 1, while $\frac{9}{15}$ is less than 1.

Example 1.2.28

Simplify $\frac{3}{5} + \frac{1}{4}$

Solution.

$$\frac{3}{5} + \frac{1}{4} = \frac{12}{20} + \frac{5}{20} = \frac{12+5}{20} = \frac{17}{20}$$

The third expression may be skipped if you feel comfortable, but this is a nice technique to practice in smaller situations (beacuse in some larger situations, it may be harder for us to simplify the addition of the numerators in our head). The purpose of showing the second expression $\frac{12}{20} + \frac{5}{20}$ being equal to the third expression $\frac{12+5}{20}$ is also helpful in seeing the Adding Fractions formula apply as literally as possible.

Example 1.2.29

Simplify $\frac{3}{10} + \frac{4}{15}$

Solution.

$$\frac{3}{10}+\frac{4}{15}=\frac{9}{30}+\frac{8}{30}=\frac{9+8}{30}=\frac{17}{30}$$

Because we are doing algebra, our fractions may have variables in them. The Adding Fractions formula still applies, and we still need a common denominator to simplify the addition of fractions.

Example 1.2.30

Simplify $\frac{2x}{3}+\frac{5x}{4}$

Solution.

$$\frac{2x}{3}+\frac{5x}{4}=\frac{8x}{12}+\frac{15x}{12}=\frac{8x+15x}{12}=\frac{23x}{12}$$

Example 1.2.31

Simplify $\frac{3}{2x^2y}+\frac{5}{4x}$

Solution. Recall that x^2 means $x \cdot x$. To achieve a common denominator of $4x^2y$, let's multiply both the top and bottom of the first fraction by 2, and multiply both the top and bottom of the second fraction by xy.

$$\frac{3}{2x^2y}+\frac{5}{4x}=\frac{6}{4x^2y}+\frac{5xy}{4x^2y}=\frac{6+5xy}{4x^2y}$$

Note that our final answer is $\frac{6+5xy}{4x^2y}$, and there is nothing that will cancel: the denominator is a product (the result of multiplication) so its pieces (called factors) are *eligible* to be cancelled, but the numerator is not a product, and so we cannot cancel any common factors on top and bottom (since the top is not written as a product).

Example 1.2.32

Simplify $\frac{4a}{3bcd}+\frac{2b}{5acd^2}$

Solution. Recall that d^2 means $d \cdot d$. To achieve a common denominator of $15abcd^2$, let's multiply both the top and bottom of the first fraction by $5d$, and multiply both the top and bottom of the second fraction by $3a$. (Be sure to read $15abcd^2$ using the Order of Operations: the exponent does not apply to all variables, just to d.)

$$\frac{4a}{3bcd}+\frac{2b}{5acd^2}=\frac{20ad}{15abcd^2}+\frac{6b^2}{15abcd^2}=\frac{20ad+6b^2}{15abcd^2}$$

How should we add a fraction and a non-fraction together? Anything that presents as a non-fraction can be written as a fraction with denominator 1.

How to add a fraction and a non-fraction.

Turn the non-fraction into a fraction with denominator 1. Then use the Adding Fractions formula, noting that this formula requires a common denominator.

Example 1.2.33

Simplify $2 + \frac{3}{5}$

Solution. We rewrite 2 as $\frac{2}{1}$. Then we can use the Adding Fractions formula, noting that we need a common denominator.

$$2 + \frac{3}{5} = \frac{2}{1} + \frac{3}{5} = \frac{10}{5} + \frac{3}{5} = \frac{10+3}{5} = \frac{13}{5}.$$

Example 1.2.34

Simplify $\frac{2}{7} + 3$

Solution. We rewrite 3 as $\frac{3}{1}$.

$$\frac{2}{7} + 3 = \frac{2}{7} + \frac{3}{1} = \frac{2}{7} + \frac{21}{7} = \frac{2+21}{7} = \frac{23}{7}.$$

Like adding fractions, subtracting requires a common denominator:

Subtracting Fractions.

Simplifying the subtraction of fractions requires having a *common denominator*.

$$\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}.$$

Example 1.2.35

Simplify $\frac{5}{6} - \frac{2}{9}$

Solution.

$$\frac{5}{6} - \frac{2}{9} = \frac{15}{18} - \frac{4}{18} = \frac{15-4}{18} = \frac{11}{18}$$

The third expression $\frac{15-4}{18}$ may be skipped if you feel comfortable, but we specifically included this step because going from the second expression $\frac{15}{18} - \frac{4}{18}$ to the third expression $\frac{15-4}{18}$ highlights the role of the *Subtracting Fractions* formula.

Example 1.2.36

Simplify $\frac{5}{12a} - \frac{3}{8ab}$

Solution.

$$\frac{5}{12a} - \frac{3}{8ab} = \frac{10b}{24ab} - \frac{9a}{24ab} = \frac{10b - 9a}{24ab}$$

In this example, to achieve a common denominator of $24ab$, the top and bottom of the first fraction both got multiplied by $2b$, while the top and bottom of the second fraction both got multiplied by $3a$.

Note that nothing cancels in $\frac{10b-9a}{24ab}$, because the denominator is a product (the result of multiplication) so its pieces (called factors) are *eligible* to be cancelled, but the numerator is not a product, and so we cannot cancel any common factors on top and bottom (since the top is not written as a product).

Example 1.2.37

Simplify $\frac{4a}{3bc} - \frac{2b}{5ac}$

Solution.

$$\frac{4a}{3bc} - \frac{2b}{5ac} = \frac{20a^2}{15abc} - \frac{6b^2}{15abc} = \frac{20a^2 - 6b^2}{15abc}$$

In this example, to achieve a common denominator of $15abc$, the top and bottom of the first fraction both got multiplied by $5a$, while the top and bottom of the second fraction both got multiplied by $3b$.

Subtraction involving a fraction and a non-fraction.

Turn the non-fraction into a fraction with denominator 1. Then use the Subtracting Fractions formula, noting that this formula requires a common denominator.

Example 1.2.38

Simplify $8 - \frac{3}{10}$

Solution. Rewrite 8 as $\frac{8}{1}$.

$$8 - \frac{3}{10} = \frac{8}{1} - \frac{3}{10} = \frac{80}{10} - \frac{3}{10} = \frac{80 - 3}{10} = \frac{77}{10}.$$

1.2.4 Multiplying fractions

The process for multiplying fractions feels really different from adding or subtracting fractions. To be convinced that the process we will describe in a moment works, we'll go back to the geometric interpretation of multiplication. Recall that $a \cdot b$ can be represented as the area of a rectangle with height a and width b.

Example 1.2.39

Represent 2×3 geometrically.

Solution.

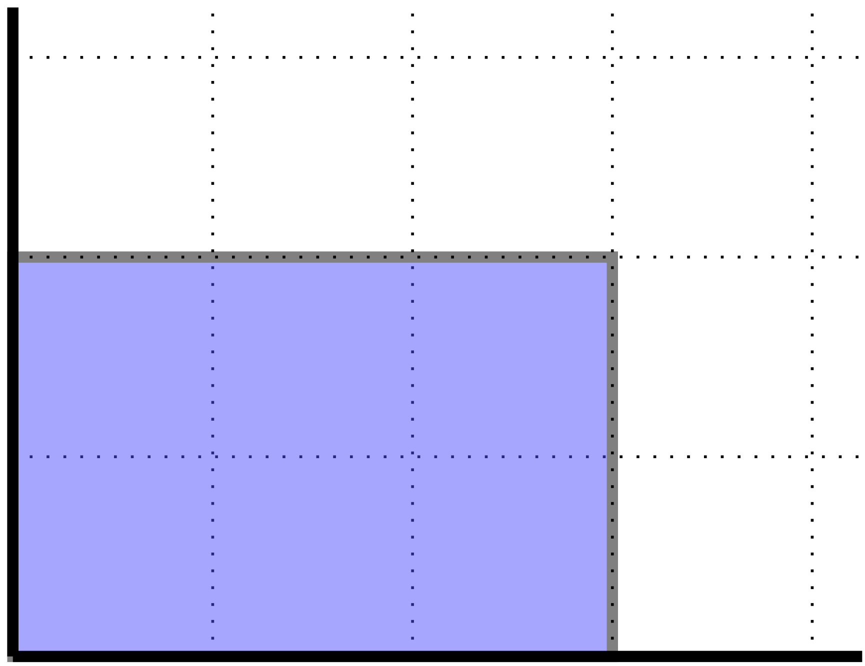

Figure 1.2.40 A picture of 2×3

The dotted lines in the picture above help us see that the rectangle has a height of 2 and a width of 3. We can be really convinced that the resulting area is 6 because we can notice the rectangle is made up of 6 unit squares. I understand that it probably still feels a little funny to represent multiplication as an area, but this is a really useful way to think about multiplication.

Example 1.2.41

Represent $\frac{1}{4} \cdot \frac{2}{3}$ geometrically. Use the picture to determine the value of $\frac{1}{4} \cdot \frac{2}{3}$.

Solution. It can be a little more challenging to represent the area. To standardize things, let's copy (to scale) one of the unit squares from the previous example. Since the first factor is $\frac{1}{4}$, we need a height of $\frac{1}{4}$. The entire square that we copied has a height of 1 so to get a height of $\frac{1}{4}$, we can divide the height of the square into 4 equal parts (achieved by creating 3> equally-spaced apart cuts in the interior of this line segment) and take 1 of those parts. Similarly, for a width of $\frac{2}{3}$, we can divide the width of the square into 3 equal parts (achieved by creating 2 equally-spaced apart cuts in the interior of this line segment) and take 2 of those parts.

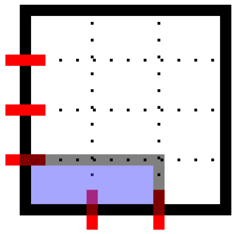

Figure 1.2.42 Picture of $\frac{1}{4} \cdot \frac{2}{3}$

By going with what we have said about the geometric interpretation of multiplication, $\frac{1}{4} \cdot \frac{2}{3}$ is the area of the blue rectangle. The portion shaded in blue is definitely less than 1 because shading the entire unit square would have had area 1. How can we determine the area of the blue rectangle exactly, though? Surprisingly, we can apply the geometric interpretation of a fraction! Recall that $\frac{a}{b}$ means to take a whole pizza and cut it into b equal slices and take a of those slices, but there was no requirement that the pizza had to be a circle. So, we can think of the entire unit square as a whole pizza. To get the area of the blue rectangle, we can see all of the dotted lines as cuts that divide the pizza into equal slices. From our starting square and uncut pizza, the cuts end up creating 12 equal slices (arranged in 4 rows and 3 columns). Then the blue rectange (representing the eaten part of the pizza) is made up of 2 of those 12 equal slices, so the area of the blue rectangle is $\frac{2}{12}$ which we can reduce to $\frac{1}{6}$.

This is a profound first example that uses a lot of ideas, so be patient with yourself! We used the geometric interpretation of multiplication to *create* the shaded region in the first place. Once we have the region we need the area of, we stopped thinking about the geometric interpretation of multiplication and instead thought about the geometric representation of a fraction. Imagine someone using a pizza cutter to cut horizontally along 3 lines, and to cut vertically in 2 lines, as shown as dotted lines in Figure 1.2.42. (The figure indicates in red the places where the pizza cutter began to cut the crust.) Out of 12 equal-sized pieces of pizza, the area of the blue rectangle is made up of 2 of those pieces, so the area of the blue rectangle is $\frac{2}{12}$ which reduces to $\frac{1}{6}$.

Example 1.2.43

Represent $\frac{1}{5} \cdot \frac{3}{4}$ geometrically. Use the picture to determine the value of $\frac{1}{5} \cdot \frac{3}{4}$.

Solution. Starting from the exact same size unit square used in the last examples, we draw inside a rectangle with height $\frac{1}{5}$ and width $\frac{3}{4}$. A height of $\frac{1}{5}$ is achieved by dividing square's height of 1 into 5 equal parts (achieved by creating 4 equally-spaced apart cuts in the interior of this line segment) and taking 1 of those parts. A width of $\frac{3}{4}$ is achieved by dividing the square's width of 1 into 4 equal parts (achieved by creating 3 equally-spaced apart cuts in the interior of this line segment) and taking 3 of those parts.

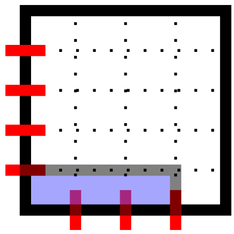

Figure 1.2.44 Picture of $\frac{1}{5} \cdot \frac{3}{4}$

The area of the blue shaded rectangle answers the question of what $\frac{1}{5} \cdot \frac{3}{4}$ is. The dotted lines represent cuts that divide the pizza into 20 equal slices (arranged in 5 rows and 4 columns). The shaded part represents taking 3 of those slices. Thus, the product is $\frac{3}{20}$.

Example 1.2.45

Represent $\frac{2}{5} \cdot \frac{3}{7}$ geometrically. Use the picture to determine the value of $\frac{2}{5} \cdot \frac{3}{7}$.

Solution. Starting from the exact same size unit square used in the last examples, we draw inside a rectangle with height $\frac{2}{5}$ and width $\frac{3}{7}$. A height of $\frac{2}{5}$ is achieved by dividing square's height of 1 into 5 equal parts (achieved by creating 4 equally-spaced apart cuts in the interior of this line segment) and taking 2 of those parts. A width of $\frac{3}{7}$ is achieved by dividing the square's width of 1 into 7 equal parts (achieved by creating 6 equally-spaced apart cuts in the interior of this line segment) and taking 3 of those parts.

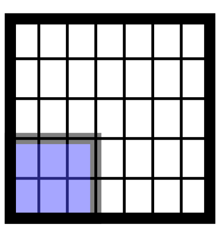

Figure 1.2.46 Picture of $\frac{2}{5} \cdot \frac{3}{7}$

The area of the blue shaded rectangle answers the question of what $\frac{2}{5} \cdot \frac{3}{7}$ is. The dotted lines represent cuts that divide the pizza into 35 equal slices (arranged in 5 rows and 7 columns). The shaded part represents taking 6 of those slices. Thus, the product is $\frac{6}{35}$.

The last example probably felt a little annoying, but I wanted us to explore the product of two fractions where neither fraction had 1 in the numerator, so that we can see if there's a pattern. (Can you guess what the pattern is?) In fact, I thought I'd show the example we just did, so that you can try a slightly less annoying example. If you have a guess for what the pattern is, can you confirm it with drawing a picture in the next example. If you don't have a guess yet, that's totally okay, and I encourage you to try making a drawing for the next example (so that you have another example to look at for the pattern).

Try it 1.2.47

Represent $\frac{3}{4} \cdot \frac{2}{5}$ geometrically. Use the picture to determine the value of $\frac{3}{4} \cdot \frac{2}{5}$.

Did you try this exercise? What did you find? Let's go back , where we saw that $\frac{2}{5} \cdot \frac{3}{7} = \frac{6}{35}$. Let's examine the resulting fraction $\frac{6}{35}$.

1. The denominator was the total number of pizza slices. Since the pizza was cut in such a way that there were 5 rows and 7 columns, there were a total of 35 slices. This number happens to be the product of the denominators of the two fractions we were originally given: $5 \cdot 7 = 35$,

2. The numerator was the number of pizza slices that were eaten, shown in the figure shaded in blue. The shaded rectangle is an arrangement of pizza slices in 2 rows and 3 columns, so there were a total of 6 slices eaten. This number happens to be the product of the numerators of the two fractions we were originally given: $2 \cdot 3 = 6$.

Did you find the same thing in Try it 1.2.47? Could you describe the numerator and denominator of the product by talking about the numerators and denominators of the two fractions we started with? If you didn't make a drawing, I encourage you to go back and try it. The point of drawing this a couple times is to be convinced that there is a pattern, and once you know that the pattern is always there (and even more awesomely, can see why it works by trying it hands on), then it won't be too surprising for you when we present the following procedure for simplifying the product of fractions.

Multiplying Fractions.

To simplify the multiplication of fractions, we use

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

Notice that this formula does not require a common denominator. This is a big difference from the Adding Fractions and Subtracting Fractions formulas. Even when we looked at , where we saw that $\frac{2}{5} \cdot \frac{3}{7} = \frac{6}{35}$, we didn't need a common denominator: the two fractions that we started with had denominators 5 and 7, which are different. The formula $\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$ communicates that common denominators are not required by having different letters in the denominators for the two fractions on the left side of the equation. The right side of the equation $\frac{a \cdot c}{b \cdot d}$ communicates that we just "multiply straight across". This is the opposite of what we saw in the Adding Fractions formula, where we were specifically warned not to "add straight across". If you take a quick minute to check that this formula applied to all the examples where we drew pictures, then it'll feel a lot more comfortable going through the next few examples where we don't draw pictures and just multiply straight across.

Example 1.2.48

Simplify $\frac{3}{7} \cdot \frac{10}{11}$

Solution. $\frac{3}{7} \cdot \frac{10}{11} = \frac{3 \cdot 10}{7 \cdot 11} = \frac{30}{77}$

In the example above, we showed the second expression $\frac{3 \cdot 10}{7 \cdot 11}$ becauuse going from the first expression to the second expression literally highlights the use of the Multiplying Fractions formula. If you feel comfortable, you can skip writing the second expression in future examples, going from the first expression to the third expression.

Example 1.2.49

Simplify $\frac{5}{9} \cdot \frac{4}{7}$

Solution 1. $\frac{5}{9} \cdot \frac{4}{7} = \frac{5 \cdot 4}{9 \cdot 7} = \frac{20}{63}$

Solution 2. If we felt comfortable skipping the second expression, our work would look like this: $\frac{5}{9} \cdot \frac{4}{7} = \frac{20}{63}$

How should we multiply a fraction and a non-fraction together? Will the technique we used for adding and subtracting a fraction and a non-fraction work here too? Yes! Anything that presents as a non-fraction can be written as a fraction with denominator 1.

Multiplying a Fraction and a Non-Fraction.

To multiply a fraction and a non-fraction, we can rewrite the non-fraction as a fraction with a denominator of 1. Then we can use the Multiplying Fractions formula.

Example 1.2.50

Simplify $2 \cdot \frac{3}{5}$

Solution. We rewrite 2 as $\frac{2}{1}$.

$$2 \cdot \frac{3}{5} = \frac{2}{1} \cdot \frac{3}{5} = \frac{2 \cdot 3}{1 \cdot 5} = \frac{6}{5}$$

Let's practice additional examples that illustrate the Multiplying Fractions formula.

Example 1.2.51

Simplify $\frac{4}{9} \cdot \frac{3}{8}$

Solution. $\frac{4}{9} \cdot \frac{3}{8} = \frac{12}{72} = \frac{1}{6}$

In this example, we reduced the fraction $\frac{12}{72}$ to $\frac{1}{6}$ in the final step. (Remember that we can only cancel common *factors*, not common *terms*. We canceled a factor of 12 from the numerator and denominator.)

Sometimes, we can simplify a multiplication of fractions by canceling common factors before we simplify the straight across multiplication. That is, we can *indicate* that multiplication is happening without *simplifying* the multiplication:

Example 1.2.52

Simplify $\frac{4}{9} \cdot \frac{3}{8}$

Solution. $\frac{4}{9} \cdot \frac{3}{8} = \frac{4 \cdot 3}{9 \cdot 8}$

Now we can see that there is a common factor of 4 in the numerator and denominator, so we can cancel that common factor:

$\frac{4 \cdot 3}{9 \cdot 8} = \frac{\cancel{4} \cdot 3}{9 \cdot \cancel{8}} = \frac{3}{9 \cdot 2} = \frac{3}{18} = \frac{1}{6}$

This is the same answer we got in the previous example, but this time we canceled a common factor before doing the multiplication. This technique can be really helpful especially when simplifying the multiplications in the numerator and/or denominator would lead to large numbers: we avoid needing to do large-number multiplication, and also, it can be easier to spot common factors before multiplying.

Example 1.2.53

Simplify $\frac{5}{12} \cdot \frac{8}{15}$

Solution. $\frac{5}{12} \cdot \frac{8}{15} = \frac{5 \cdot 8}{12 \cdot 15}$

Now we can see that there is a common factor of 4 in the numerator and denominator as well as a common factor of 5 in the numerator and denominator (or instead of seeing the 4s and 5s individuall

$\frac{5\cdot 8}{12\cdot 15} = \frac{\cancel{4}\cdot\cancel{5}\cdot 2}{\cancel{4}\cdot\cancel{5}\cdot 9} = \frac{2}{9}$

Please note that you don't have to do it this way, but it can be convenient. If we write that there is a multiplication happening but do not simplify multiplication, it might just be easier to spot common factors. Cancelling early on means that when we eventually do simplify the multiplication, we will be working with smaller numbers.

Example 1.2.54

Simplify $\frac{20}{27} \cdot \frac{9}{14}$

Solution 1. $\frac{20}{27} \cdot \frac{9}{14} = \frac{20\cdot 9}{27\cdot 14} = \frac{180}{378} = \frac{10}{21}$

It takes a bit of work to simplify the multiplications in the numerator and denominator to obtain $\frac{180}{378}$ in the first place. Then, due to the numbers being large, it takes considerable effort to reduce this fraction to its equal value $\frac{10}{21}$. This solution is completely valid, but the second solution below shows how we can avoid the large-number multiplications by canceling common factors before simplifying the multiplication.

Solution 2. $\frac{20}{27} \cdot \frac{9}{14} = \frac{20\cdot 9}{27\cdot 14}$

The numerator and denominator both have factors of 9 and 2, so we can cancel those common factors:

$\frac{20\cdot 9}{27\cdot 14} = \frac{\cancel{9}\cdot\cancel{2}\cdot 10}{\cancel{9}\cdot 3\cdot\cancel{2}\cdot 7} = \frac{10}{21}$

Example 1.2.55

Simplify $\frac{3ab}{10c} \cdot \frac{55bc}{21a^2}$

Solution. First we indicate the multiplication without simplifying:

$\frac{3ab}{10c} \cdot \frac{55bc}{21a^2} = \frac{3ab\cdot 55bc}{10c\cdot 21a^2} = \frac{3ab\cdot 55bc}{10c\cdot 21\cdot a\cdot a}$

Above, we rewrote a^2 as $a \cdot a$. The factors that appear in both the numerator and denominator are 5 and 3 and a and c.

$\frac{3ab\cdot 55bc}{10c\cdot 21\cdot a\cdot a} = \frac{\cancel{3}\cdot b\cdot\cancel{5}\cdot 11\cdot b\cdot\cancel{c}}{\cancel{5}\cdot 2\cdot\cancel{3}\cdot 7\cdot\cancel{a}\cdot a} = \frac{11\cdot b\cdot b}{14a} = \frac{11b^2}{14a}$ where we rewrote $b \cdot b$ as b^2.

Warning 1.2.56 Please note that when multiplying fractions, we do not need a common denominator: sometimes people get overly cautious and try to treat multiplying fractions like adding fractions. Let's look at an example and see what the common error is:

- When asked to simplify $\frac{4}{9} \cdot \frac{5}{9}$ it is tempting to think about common denominators (because we notice that both fractions already have the same denominator of 9) but it is incorrect to state the result of simplying this multiplication as $\frac{20}{9}$.

- Before talking about what we *should* do and ignoring our work above, let's talk about the work above, to see if we can take the experience and foundation that we've built to question the reasonableness of our answer: Notice that the numerator is larger than the denominator in $\frac{20}{9}$, so this fraction is greater than 1, but both of the fractions we started with were less than 1, so it doesn't make sense that their product would be greater than 1. In fact, the pizza diagram for $\frac{4}{9} \cdot \frac{5}{9}$ would take just *some* of the slices from a square pizza that was cut into a 9-by-9 grid of 81 slices, so the resulting fraction should be less than 1.

- The correct simplification is $\frac{4}{9} \cdot \frac{5}{9} = \frac{4 \cdot 5}{9 \cdot 9} = \frac{20}{81}$. Notice that the denominator of the result is 81, not 9. The denominator of the result is the product of the denominators of the two fractions we started with: $9 \cdot 9 = 81$, even though the two denominators that we started with were the same. (This is different from adding fractions, where the denominator of the result was the common denominator.) The fact that we had common denominators is actually a distraction, and we should ignore it.

- Here is a silghtly different example: $\frac{4}{9} \cdot \frac{5}{10}$. While this is a different problem, because the only change was by slightly changing the denominator of the second fraction, it seems reasonable that the final answer to this question would be close to the final answer of the previous question. Here, the denominators are different, so there is no distraction. Following our Multiplying Fractions formula, we have $\frac{4}{9} \cdot \frac{5}{10} = \frac{20}{90}$ by multiplying straight across. Use a calculator to verify that $\frac{20}{90}$ is close to $\frac{20}{81}$, but far from $\frac{20}{9}$.

The takeaway is that when multiplying fractions just multiply straight across whether we have a common denominator or not!

1.2.5 Dividing fractions

Dividing by a fraction is the same as multiplying by its reciprocal.

Dividing Fractions formula.

To divide by a fraction, multiply by its reciprocal and follow the Multiplying Fractions formula to simplify the multiplication of fractions.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

Example 1.2.57

Simplify $\frac{2}{3} \div \frac{5}{7}$

Solution. $\frac{2}{3} \div \frac{5}{7} = \frac{2}{3} \cdot \frac{7}{5} = \frac{14}{15}$

Example 1.2.58

Simplify $\frac{\frac{4}{9}}{\frac{2}{3}}$

Solution. Although the question didn't write the division symbol $\div$, the fraction bar indicates division. (So we could rewrite the problem as $\frac{4}{9} \div \frac{2}{3}$, and we'll say this is asking the same question.) However, to practice this new format, let's leave the original question in this format and work from there.

$$\frac{\frac{4}{9}}{\frac{2}{3}} = \frac{4}{9} \cdot \frac{3}{2} = \frac{12}{18} = \frac{2}{3}.$$

From the second-to-last expression to the last expression, we reduced the fraction.

Warning 1.2.59 It is often tempting to see a question in this format $\frac{\frac{4}{9}}{\frac{2}{3}}$ and in the desire to multiply by the reciprocal, something unintentional remains in the writing. For example, looking to the previous computation, when starting with $\frac{\frac{4}{9}}{\frac{2}{3}}$ the next expression shouldn't read $\frac{\frac{4}{9} \cdot \frac{3}{2}}{\frac{2}{3}}$. The point is that multiplying by $\frac{3}{2}$ is what *replaces* dividing by $\frac{2}{3}$, so we should either have $\frac{2}{3}$ in the denominator or have the multiplication by $\frac{3}{2}$, but not both.

The analogy in the question before this one would be like if someone turned $\frac{2}{3} \div \frac{5}{7}$ into $\frac{2}{3} \div \frac{5}{7} \cdot \frac{7}{5}$ instead of $\frac{2}{3} \cdot \frac{7}{5}$. Multiplying by $\frac{7}{5}$ is what *replaces* dividing by $\frac{5}{7}$, so we should either divide by $\frac{5}{7}$ or multiply by $\frac{7}{5}$, but not both.

Example 1.2.60

Simplify $\frac{\frac{3}{4ac}}{\frac{5b}{6a}}$

Solution.

$$\frac{\frac{3}{4ac}}{\frac{5b}{6a}} = \frac{3}{4ac} \cdot \frac{6a}{5b} = \frac{18a}{20abc} = \frac{9}{10bc}$$

How would we divide between a fraction and a non-fraction? As usual, turn the non-fraction into a fraction with denominator 1.

Example 1.2.61

Simplify $2 \div \frac{3}{5}$.

Solution. We rewrite 2 as $\frac{2}{1}$. Then:

$$2 \div \frac{3}{5} = \frac{2}{1} \div \frac{3}{5} = \frac{2}{1} \cdot \frac{5}{3} = \frac{10}{3}$$

1.2.6 Additional practice

Example 1.2.62

Simplify $\frac{3}{4} + \frac{2}{5} \cdot \frac{7}{8}$

Solution.

$$\frac{3}{4} + \frac{2}{5} \cdot \frac{7}{8} = \frac{3}{4} + \frac{14}{40} = \frac{3}{4} + \frac{7}{20} = \frac{15}{20} + \frac{7}{20} = \frac{22}{20} = \frac{11}{10}.$$

Note that we had to simplify multiplication before simplifying addition. It is easy to fall into the trap of simplifying addition (that is, working on simplifying $\frac{3}{4} + \frac{2}{5}$) first, because we see the addition written first. However, we still need to read expressions based on the Order of Operations, even if the expression contains fractions.

Example 1.2.63

Simplify $\frac{2}{3} - \frac{6}{7} \div \frac{5}{2}$

Solution. The division must be simplified before the subtraction. We turn dividing by $\frac{5}{2}$ into multiplying by $\frac{2}{5}$ instead.

$$\frac{2}{3} - \frac{6}{7} \div \frac{5}{2} = \frac{2}{3} - \frac{6}{7} \cdot \frac{2}{5} = \frac{2}{3} - \frac{12}{35} = \frac{70}{105} - \frac{36}{105} = \frac{34}{105}.$$

Example 1.2.64

Simplify $\frac{2}{3x} - \frac{6y}{7} \cdot \frac{1}{2x}$.

Solution.

$$\frac{2}{3x} - \frac{6y}{7} \cdot \frac{1}{2x} = \frac{2}{3x} - \frac{6y \cdot 1}{7 \cdot 2x} = \frac{2}{3x} - \frac{6y}{14x} = \frac{2}{3x} - \frac{3y}{7x} = \frac{14 - 9y}{21x}.$$

Note that we cannot cancel part of the 14 with part of the 21 because 14 is not a factor of the numerator. Similarly, we cannot cancel part of the 9 with part of the 21 because 9 is not a factor of the numerator.

Example 1.2.65

Simplify $\frac{2x}{3y} + \frac{\frac{3x}{4y}}{\frac{5y}{6x}}$

Solution. We can simplify the division of fractions before simplifying addition:

$$\frac{2x}{3y} + \frac{\frac{3x}{4y}}{\frac{5y}{6x}} = \frac{2x}{3y} + \frac{3x \cdot 6x}{4y \cdot 5y} = \frac{2x}{3y} + \frac{18x^2}{20y^2} = \frac{2x}{3y} + \frac{9x^2}{10y^2}.$$

Now we need a common denominator to simplify addition of fractions:

$$\frac{2x}{3y} + \frac{9x^2}{10y^2} = \frac{2x \cdot 10y}{3y \cdot 10y} + \frac{9x^2 \cdot 3y}{10y^2 \cdot 3y} = \frac{20xy}{30y^2} + \frac{27x^2y}{30y^2} = \frac{20xy + 27x^2y}{30y^2}.$$

Finally, we can factor out y in the numerator, and having factors of y in both the numerator and denominator allows us to cancel y:

$$\frac{20xy + 27x^2y}{30y^2} = \frac{y(20x + 27x^2)}{30y^2} = \frac{20x + 27x^2}{30y}.$$

1.2.7 Clarifications and details

Let's clarify some details about fractions. It will be easier to discuss this using a concrete example, but the discussion here applies in general. What we discuss will lead to important clarifications and expectations. Along the way, we discuss convenient places to stop working on a problem (it's always good to know when to *stop* and not work further than you have to!), because some formats of writing will be easier to work with in algebra than other formats.

Suppose somone wrote $2\frac{3}{5}$. Then:

- Someone might have written $2\frac{3}{5}$ with the intent of writing a **mixed fraction**. The *mixed* part of the name refers to the fact that there is a non-fraction part (2) and a fraction part ($\frac{3}{5}$) mixed together. Using the standard procedure for converting a mixed fraction into an improper fraction, the value of $2\frac{3}{5}$ is $\frac{2 \cdot 5 + 3}{5}$ which simplifies to $\frac{13}{5}$.

  This is a good time to mention that writing $2\frac{3}{5}$ to mean a mixed fraction is actually shorthand for $2 + \frac{3}{5}$. In fact, in Example 1.2.33 we had simplified $2 + \frac{3}{5}$ to get $\frac{13}{5}$.

- An entirely different view is possible. Someone might have written $2\frac{3}{5}$ to mean $2 \cdot \frac{3}{5}$, since a missing operation symbol is actually a hidden multiplication sign. In Example 1.2.50, we simplified $2 \cdot \frac{3}{5}$ to get $\frac{6}{5}$. Please note that $\frac{6}{5}$ is not the same as $\frac{13}{5}$. The two interpretations that we already have lead to different results: the mixed fraction intent leads to $\frac{13}{5}$ while treating a missing operation symbol as multiplication leads to $\frac{6}{5}$.

- We should bring up a third possibility, though what we are about to bring up is a common error of writing. Suppose we were in a situation where someone needs to start with the fraction $\frac{3}{5}$ and multiply the top and bottom by 2. (A situation where this might happen is when we are tasked with simplifying $\frac{3}{5} + \frac{7}{10}$, and we'd need a common denominator to simplify the addition of fractions.) In a situation where someone needs to start with the fraction $\frac{3}{5}$ and multiply the top and bottom by 2, they might just write a single 2 next to the fraction, which would look like this: $2\frac{3}{5}$. However, this is not a correct way to write what the writer intends. With this incorrect writing the writer would usually recover and correct for this in the next step by writing $\frac{6}{10}$ the result of actually multiplying by 2 on top and bottom. You can use a calculator to check that $\frac{6}{10}$ is not equal to either $\frac{13}{5}$ or $\frac{6}{5}$. In fact, $\frac{6}{10}$ is equal to $\frac{3}{5}$, which makes sense because of what the Equal Fractions formula tells us.

  When the desire is to multiply by two on top and on bottom, there are two ways to do this that are clear and unambiguous:

  1. Write the "times 2" both on top *and* on bottom: $\frac{3}{5} = \frac{2 \cdot 3}{2 \cdot 5} = \frac{6}{10}$.
  2. Write *neither* the 2 on top *nor* the 2 on bottom, but perform and simplify these actions nonetheless: $\frac{3}{5} = \frac{6}{10}$

  But what we *really* must avoid doing is writing a *single* 2 to mean that we are multiplying by 2 on top and bottom. So writing $2\frac{3}{5}$ cannot mean for us that we are multiplying by 2 on top and on bottom, even if there's a lot of spacing between the 2 and the fraction. If you wish to write what you're multiplying by, just be sure to write two copies: one on top and one on bottom.

Maybe it's shocking to see that the same excerpt of writing $2\frac{3}{5}$ can be interpreted two different ways leading to two different numbers. This discussion leads to some important ideas. First, let's present an expectation regarding our final answers for fractions:

Expectation for Fraction Format.

We expect our final answers for fractions to be in improper fraction form whenever possible. Unless specifically requested by the question, we will never present an answer as a mixed fraction. Whenever we write a mixed fraction, we should *actually mention* that our writing should be interpreted as a mixed fraction.

You may be more used to mixed fractions and feel offended that the expectation avoids mixed fractions and favors improper fractions instead. But improper fractions are easier to work with in algebra, because it is hard to directly add, subtract, multiply, or divide mixed fractions. (In fact, when given a mixed fraction and a second mixed fraction, to perform any operation, we first need to convert both mixed fractions into improper fractions.) Therefore, it's actually convenient to leave any fractions as improper fractions in their final answers.

Example 1.2.66

Simplify $2\frac{3}{5} \cdot 3\frac{7}{10}$ where we interpret $2\frac{3}{5}$ and $3\frac{7}{10}$ as mixed fractions.

Solution. We rewrite the first mixed fraction $2\frac{3}{5}$ as $\frac{13}{5}$. We rewrite the second mixed fraction $3\frac{7}{10}$ as $\frac{37}{10}$. Then, the product is

$$\frac{13}{5} \cdot \frac{37}{10} = \frac{481}{50}.$$

By going through this example, I hope you really saw that it's preferrable to work with improper fractions anyway. In this question, we only had the step of completing one opertaion, but in a larger problem, we might need to take the result and do something further. It would be more convenient to do any further task working the improper fraction $\frac{481}{50}$, so that's why we don't really bother converting this into a mixed fraction.

Note 1.2.67 Perhaps the history of calling $\frac{481}{50}$ an **improper fraction** is what makes us a little uneasy. We use this language here (as opposed to the fraction $\frac{47}{50}$) because the numerator is larger than the denominator in $\frac{481}{50}$. I guess I'd call $\frac{x}{y+3}$ an "improper fraction" only because I wouldn't call $\frac{x}{y+3}$ a mixed fraction, but because there are variables, I don't really know if x is larger than $y + 3$ or not. However, $\frac{x}{y+3}$ behaves like it needs to, and we can apply our fraction operations as described. For example,

$$\frac{x}{y+3} + \frac{z-6}{y+3} = \frac{x+z-6}{y+3}.$$

Just like improper fractions are easier to work with than mixed fractions, improper fractions are are generally easier to work with than decimal represtations. Therefore, it is often more convenient for future steps to leave $\frac{9}{8}$ as is instead of presenting this as 1.125, and similarly, it is more convenient for us (and less work!) to leave $\frac{3}{8}$ as it is instead of presenting this as 0.375.

Well then, why have mixed fractions or decimals in the first place?

In everyday settings, people are used to seeing mixed fractions. Therefore, at the very end of an applied problem, when we are done and we are certain that the value will not be needed in further work, it would be reasonable to state a final answer (in this limited situation) as a mixed fraction.

The role of decimals is somewhat similar: when you have a final answer that is an improper fraction but you need to know a "ballpark figure" representation of that number, a decimal is helpful. For example, I don't really have a good sense of the value of $\frac{2711348695}{182638791}$, but I can put this into a calculator to see the value is close to 14.8454152601, a decimal representation.

Changing gears, there's one thing that we haven't mentioned yet that's worth clarifying: The top and bottom of a fraction are each in what I'd like to call hidden parentheses.

Fractions have hidden parentheses.

The top and bottom of a fraction are each in hidden parentheses..

Let's clarify what this means, and then talk about how this connects to the Order of Operations. The connection to the Order of Operations is really important in applied settings when a fraction-based expression on paper needs to be input into a calculator to get a decimal approximation.

Example 1.2.68

In the fraction $\frac{2+3}{5-1}$, the top is $2+3$ and the bottom is $5-1$. The top and bottom are each in hidden parentheses, so we can rewrite this as $\frac{(2+3)}{(5-1)}$. We can also write this as $(2+3) \div (5-1)$, but cannot write this as $2+3 \div 5-1$ due to the Order of Operations. Therefore, to enter $\frac{2+3}{5-1}$ into a calculator, type $(2+3) \div (5-1)$, including the parentheses.

Example 1.2.69

From a general chemistry class, a student got a value of $\frac{3.6 \cdot 10^{-5}}{(4.0 \cdot 10^{-2})(2.5 \cdot 10^{-2})}$ on paper. How should this be entered on a calculator?

Solution. There are two sets of parentheses in the denominator already, but we should insert a set of parentheses around the entire denominator. We should also insert a set of parentheses around the numerator. Therefore, we should enter $(3.6 \cdot 10^{-5}) \div ((4.0 \cdot 10^{-2})(2.5 \cdot 10^{-2}))$ into the calculator.

Example 1.2.70

If you deposited 100 dollars every month for 10 years into an account that ears 6% interest compounded monthly, we can compute what the balance will be at the end of 10 years by applying an equation in finance, and we would obtain

$$100 \cdot \frac{(1+\frac{0.06}{12})^{(12)(10)}-1}{\frac{0.07}{12}}$$

as the balance. How should this be input into the calculator so that we can really understand what the balance will be?

Solution. Insert the entire numerator and denominator of the fraction in parentheses. In a calculator, we would type $100 * ((1+(0.06/12))^((12)(10)) -$

1)/(0.06/12).

This fact that the entire numerator and the entire denominator are each in hidden parentheses, we can enhance and clarify what we've said about when we can cancel in fractions.

Example 1.2.71

Simplify $\frac{4(x+2)}{x+2}$

Solution. Let's draw in the hidden parentheses to surround the entire denominator.

$$\frac{4(x+2)}{x+2} = \frac{4(x+2)}{(x+2)} = \frac{4}{1} = 4$$

We canceled the factor of $(x+2)$ in the numerator and denominator

If it still feels hard to view $(x+2)$ in the denominator, that's okay! I think that means you're doing a great job of *really* digging into the idea that factors are the pieces of a multiplication (and where *is* multiplication in the denominator)? Let's rewrite a solution, noting that multiplying anything by 1 doesn't change a value, so $1 \cdot (x+2) = x+2$.

$$\frac{4(x+2)}{x+2} = \frac{4 \cdot (x+2)}{1 \cdot (x+2)} = \frac{4}{1} = 4$$

In fact, writing this level of detail explains why we have a 1 instead of a 0 in the denominator after the cancellation happened.

Example 1.2.72

Simplify $\frac{(x+8)^2}{x+8}$

Solution. Recall that squaring a thing means to multiply the thing by itself, even if that ``thing'' is longer to write, like $x+8$, so $(x+8)^2 = (x+8) \cdot (x+8)$. Let's make that replacement in the numerator, and then also surround the denominator in parentheses, making those hidden parentheses visible.

$$\frac{(x+8)^2}{x+8} = \frac{(x+8) \cdot (x+8)}{1 \cdot (x+8)} = \frac{x+8}{1} = x+8$$

In the second to last expression, which is the moment we just finished cancellation, we could have written $\frac{(x+8)}{1}$ instead of $\frac{x+8}{1}$. However, since the numerator was $(x+8)$, I wanted to practice the principle that the numerator of a fraction is always in a hidden set of parentheses, but "backwards" in a sense: just as much as we can draw in the hidden parentheses around the entire top or entire bottom, we can remove parentheses that surround the entire top or entire bottom.

Example 1.2.73

Simplify $\frac{x+8}{(x+8)}{}^{2}$

Solution.

$$\frac{x+8}{(x+8)}{}^{2} = \frac{(x+8)}{(x+8)\cdot(x+8)} = \frac{1}{(x+8)} = \frac{1}{x+8}$$

We canceled the factor of $(x+8)$ in the numerator and denominator. One new thing to note in this question: after cancellation, what's left is the factor of $(x+8)$ in the denominator, and we just removed the parentheses that surrounded the entire denominator. However, a common error that occurs right at the moment of cancellation is that it's temtping to put the factor of $(x+8)$ in the numerator. Note that after cancellation, the $(x+8)$ factor that didn't cancel was in the denominator, and shouldn't move up to the numerator.

Warning 1.2.74 After cancelling the same factor on top and bottom of a fraction, anything that remains uncanceled should stay in the same part of the fraction. Factors that are in the denominator should remain in the denominator, and shouldn't move up to the numerator. Factors that are in the numerator should remain in the numerator, and shouldn't move down to the denominator.

Example 1.2.75

Simplify $\frac{(x+3)(x+5)}{(x+3)(x+5)(x+7)}$

Solution. $\frac{(x+3)(x+5)}{(x+3)(x+5)(x+7)} = \frac{1}{(x+7)} = \frac{1}{x+7}$

1.2.8 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section.

Example 1.2.76

A baking recipe calls for $5\frac{1}{4}$ cups of flour. The recipe says that it serves 10. You are hosting a party where 35 people will attend, so you need to make $3\frac{1}{2}$ times the recipe. How many cups of flour do you need?

Solution. The original recipe needs $5\frac{1}{4}$ cups of flour, but we need to make $3\frac{1}{2}$ times the recipe. We have to multiply both of these numbers, which (at least through context) we can see are both mixed fractions. We convert the two mixed fractions into improper fractions: The mixed fraction $5\frac{1}{4}$ is $\frac{21}{4}$, and the mixed fraction $3\frac{1}{2}$ is $\frac{7}{2}$. Then, we multiply the two improper fractions:

$$\frac{21}{4} \cdot \frac{7}{2} = \frac{147}{8}.$$

Now, given that the problem is about baking, it makes a lot of sense to convert our final answer into a mixed fraction, since that's how measuring cups are typically used: We need $18\frac{3}{8}$ cups of flour, as a mixed fraction. As a reminder of what mixed fraction is, if we wanted, we could actually write this quantity unambigously by writing $18 + \frac{3}{8}$ cups of flour.

Example 1.2.77

You have a two-by-four piece of lumber that measures $2\frac{5}{8}$ feet long and another that is $1\frac{3}{4}$ feet long. You will glue the two pieces together to create one leg of a nightstand. But you'll need to go to the store to make three other legs of the same height. How long is each leg?

Solution. Because we are gluing one piece of lumber to another, we need to add their measurements. From context, both numbers provided are mixed fractions, so we convert them: The mixed fraction $2\frac{5}{8}$ is $\frac{21}{8}$, and the mixed fraction $1\frac{3}{4}$ is $\frac{7}{4}$. Then, we add the two improper fractions:

$$\frac{21}{8} + \frac{7}{4} = \frac{21}{8} + \frac{14}{8} = \frac{35}{8}.$$

Since we needed to simplify the addition of fractions, we first got both fractions to have a common denominator of 8, which we achieved by leaving the first fraction alone, and multiplying the numerator and denominator of the second fraction by 2. Each leg will be $\frac{35}{8}$ feet long. Now, given that the problem is about measuring lumber, it makes a lot of sense to convert our final answer into a mixed fraction, since that's how lumber is typically measured: Each leg will be $4\frac{3}{8}$ feet long, as a mixed fraction.

Example 1.2.78

A swimming pool is being filled at a rate of $8\frac{1}{4}$ gallons per hour. How many gallons of water will be in the pool after $2\frac{1}{4}$ hours?

Solution. Because we are filling the pool at a rate of $8\frac{1}{4}$ gallons per hour, and we are filling it for $2\frac{1}{4}$ hours, we need to multiply these two numbers. From context, both numbers provided are mixed fractions, so we convert them: The mixed fraction $8\frac{1}{4}$ is $\frac{33}{4}$, and the mixed fraction $2\frac{1}{4}$ is $\frac{9}{4}$. Then, we multiply the two improper fractions:

$$\frac{33}{4} \cdot \frac{9}{4} = \frac{297}{16}.$$

The pool will have $\frac{297}{16}$ gallons of water, or $18\frac{8}{16}$ gallons of water as a mixed fraction. Note that when we simiplified $\frac{33}{4} \cdot \frac{9}{4}$, we need to multiply the two denominators together (even though they matched).

Example 1.2.79

A car is traveling at a speed of 60 miles per hour. How far will the car travel in $4\frac{1}{2}$ hours?

Solution. Because we are traveling at a rate of 60 miles per hour, and we are traveling for $4\frac{1}{2}$ hours, we need to multiply these two numbers. From context, the first number is not a mixed fraction, but the second number is a mixed fraction, so we convert it: The mixed fraction $4\frac{1}{2}$ is $\frac{9}{2}$. Now, we are trying to compute $60 \cdot \frac{9}{2}$, and to resolve the awkwardness of multiplying a non-fraction by a fraction, we will turn 60 into $\frac{60}{1}$.

$$60 \cdot \frac{9}{2} = \frac{60}{1} \cdot \frac{9}{2} = \frac{540}{2} = 270.$$

The car will travel 270 miles.

Example 1.2.80

A group of 5 friends go out to dinner. The bill comes to \$48.75. If they split the bill evenly, how much does each person pay?

Solution. We can think of 48.75 as the mixed fraction $48\frac{75}{100}$ or by reducing the fraction, we can say this is the mixed fraction $48\frac{3}{4}$, and for the sake of nice computation, as an improper fraction, this is $\frac{195}{4}$. We need the value of $\frac{195}{4}$ to be a cost shared by five people equally, so we should divide this fraction by 5. We will actually make our very next step to change the 5 into $\frac{5}{1}$.

$$\frac{195}{4} \div 5 = \frac{195}{4} \div \frac{5}{1} = \frac{195}{4} \cdot \frac{1}{5} = \frac{195}{20} = \frac{39}{4}.$$

Each person's share is $\frac{39}{4}$ dollars, and this as a mixed fraction is $9\frac{3}{4}$ dollars and as a decimal is 9.75 dollars.

Example 1.2.81

A recipe calls for $\frac{3}{4}$ cup of sugar. You have a $\frac{1}{3}$ cup measuring cup. How many $\frac{1}{3}$ cups of sugar do you need to use to get the correct amount?

Solution. We need to find the value of $\frac{3}{4} \div \frac{1}{3}$, which is the same as $\frac{3}{4} \cdot \frac{3}{1}$.

$$\frac{3}{4} \div \frac{1}{3} = \frac{3}{4} \cdot \frac{3}{1} = \frac{9}{4}.$$

We need to fill the one-third cup measuring cup that we have $\frac{9}{4}$ times, or as a mixed fraction, $2\frac{1}{4}$ times. So, to fulfill what the recipe wants, we should fill our $\frac{1}{3}$ cup measuring cup full two times, and then do our best to estimate filling this measuring cup to one-fourth full.

Example 1.2.82

A recipe calls for $\frac{3}{4}$ cup of sugar. You put in $\frac{1}{3}$ cup of sugar by accident. How much more sugar do you need to put in to get back on track in following the recipe?

Solution. Since we already put in $\frac{1}{3}$ cup of sugar, we need to find out what's left of the $\frac{3}{4}$ cup of sugar that we haven't put in yet. That is, we need to find the value of $\frac{3}{4} - \frac{1}{3}$.

$$\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}.$$

We need to put in $\frac{5}{12}$ cup of sugar to get back on track.

1.2.9 Summary

1. We can take a fraction and create a fraction of the same value by multiplying by the same quantity on the top and bottom.

2. Undoing the multiplying of the same thing on top/bottom is the cancelling of a common factor. We can only cancel common factors, not common terms. (Recall that **factor** refers to the pieces of **multiplication**.)

3. To simplify adding or subtracting fractions, we need a common denominator first.

4. To multiply fractions, multiply straight across whether we have a common denominator or not.

5. To divide two fractions, note that dividing by a fraction is the same thing as multiplying by its reciprocal.

6. We stick to improper fractions as much as possible because it is much easier to perform operations on improper fractions, and only write a mixed fraction or

decimal when an application calls for it.

7. The entire numerator of a fraction is in a set of parentheses, which can be written in explicitly, or be "hidden." Similarly, the denominator of a fraction is also in a set of parentheses, which can be written in explicitly, or be "hidden." This is important to keep in mind when entering an answer that has fractions into a calculator: ignoring this may lead to *very* different decimal value than intended.

8. The fact that both the top and bottom can each be thought of as being in a hidden set of parentheses also enhances our understanding of how we can cancel in fractions: it still remains the case that we can only cancel common *factors*, but now we can see that the *entire* top and *entire* bottom can be considered factors through this understanding of hidden parentheses.

1.2.10 EXERCISES

1. Practice rewriting fractions using the Equal Fractions formula. Pay attention to unintended meanings of notation in your work, as described in Section .

   (a) What fraction is equal to $\frac{3}{4}$ but has 20 in the denominator?
   (b) What fraction is equal to $\frac{2}{5}$ but has 20 in the denominator?
   (c) What fraction is equal to $\frac{7}{10}$ but has 100 in the denominator?
   (d) What fraction is equal to $\frac{5}{6}$ but has 12 in the denominator?
   (e) What fraction is equal to $\frac{9}{8}$ but has 32 in the denominator?
   (f) What fraction is equal to $\frac{11}{12}$ but has 36 in the denominator?
   (g) What fraction is equal to $\frac{2}{x}$ but has $3x$ in the denominator?
   (h) What fraction is equal to $\frac{5}{y}$ but has $20y$ in the denominator?
   (i) What fraction is equal to $\frac{7}{2a}$ but has $10a$ in the denominator?
   (j) What fraction is equal to $\frac{x}{3}$ but has 12 in the denominator?
   (k) What fraction is equal to $\frac{3}{2b}$ but has $14b$ in the denominator?
   (l) What fraction is equal to $\frac{y}{4z}$ but has $12z$ in the denominator?
   (m) What fraction is equal to $\frac{3a}{10}$ but has 50 in the denominator?
   (n) What fraction is equal to $\frac{5m}{6n}$ but has $30n$ in the denominator?

2. Simplify the following expressions.

   (a) $\frac{3}{2} + \frac{2}{5}$.
   (b) $\frac{5}{6} + \frac{1}{3}$.
   (c) $\frac{7}{8} - \frac{3}{8}$.
   (d) $\frac{9}{10} - \frac{2}{5}$.
   (e) $\frac{2}{3} \cdot \frac{9}{4}$.

(f) $\frac{5}{7} \cdot \frac{14}{15}$.
(g) $\frac{3}{4} \div \frac{9}{16}$.
(h) $\frac{10}{21} \div \frac{5}{14}$.
(i) $\frac{1}{2} + \frac{2}{3} \cdot \frac{3}{4}$.
(j) $\frac{5}{6} - \frac{2}{5} \cdot \frac{3}{2}$.
(k) $\frac{3}{4} + \frac{5}{8} \div \frac{5}{16}$.
(l) $\frac{7}{9} - \frac{3}{10} \cdot \frac{5}{6}$.
(m) $(\frac{3}{4} + \frac{1}{8}) \cdot \frac{2}{3}$.
(n) $(\frac{5}{6} - \frac{1}{3}) \div \frac{1}{2}$.
(o) $(\frac{7}{10} + \frac{2}{5}) \cdot \frac{3}{7}$.
(p) $(\frac{9}{8} - \frac{5}{16}) \div \frac{3}{4}$.
(q) $\frac{3}{5} + \frac{2}{3} - \frac{4}{15}$.
(r) $\frac{7}{8} \cdot \frac{2}{5} + \frac{3}{10}$.
(s) $\frac{4}{9} + \frac{2}{3} \div \frac{8}{27}$.
(t) $\frac{5}{12} - \frac{3}{16} + \frac{7}{24}$.

3. Simplify the following expressions.

(a) $\frac{x}{2} + \frac{x}{3}$.
(b) $\frac{2a}{5} - \frac{a}{10}$.
(c) $\frac{3}{x} \cdot \frac{2}{5}$.
(d) $\frac{4}{y} \div \frac{2}{3y}$.
(e) $\frac{2}{3}x + \frac{x}{4} \cdot \frac{3}{2}$.
(f) $\frac{a}{6} + \frac{3a}{4} - \frac{a}{12}$.
(g) $\frac{y}{5} - \frac{2y}{15} \cdot \frac{3}{2}$.
(h) $\frac{m}{8} + \frac{2m}{3} \div \frac{4}{9}$.
(i) $(\frac{3}{4}x + \frac{x}{8}) \cdot \frac{2}{3}$.
(j) $(\frac{a}{5} - \frac{2a}{15}) \div \frac{a}{3}$.
(k) $(\frac{y}{6} + \frac{y}{4}) \cdot \frac{3}{2}$.
(l) $(\frac{2m}{7} - \frac{m}{14}) \div \frac{m}{21}$.
(m) $\frac{x}{2} + \frac{2x}{3} - \frac{x}{6}$.
(n) $\frac{3a}{4} \cdot \frac{2}{3a} + \frac{a}{2}$.
(o) $\frac{y}{3} + \frac{2y}{9} \div \frac{y}{6}$.
(p) $\frac{m}{5} - \frac{3m}{10} + \frac{7m}{20}$.
(q) $\frac{2x}{3} + \frac{x}{4} - \frac{3x}{8} \cdot \frac{2}{3}$.
(r) $\frac{5a}{6} - (\frac{a}{4} + \frac{a}{12}) \div \frac{a}{3}$.

(s) $\frac{3y}{8} + \frac{y}{6} - \frac{y}{4} \cdot \frac{9}{2}$.

(t) $(\frac{2m}{9} + \frac{m}{6} - \frac{m}{18}) \cdot \frac{3}{2}$.

4. A runner completed $\frac{2}{3}$ of a marathon in the morning and $\frac{1}{6}$ in the evening. How much of the marathon did the runner complete in total?

5. A carpenter cuts a piece of wood $3\frac{1}{2}$ feet long from a board that is $10\frac{1}{4}$ feet long. How much of the board remains?

6. You drink $\frac{3}{4}$ of a liter of water during a workout, and then another $\frac{2}{5}$ of a liter after. How much water did you drink altogether?

7. Your health routine requires drinking $1\frac{1}{4}$ liters of protein shake a day. You started the day by having $\frac{2}{3}$ liter of protein shake. How much should you have in the rest of the day?

8. A farmer harvested $\frac{7}{8}$ of a field of corn one week and $\frac{3}{10}$ of the field the next. How much of the field is harvested in total?

9. A rope is 12 meters long. You cut off a piece that is $\frac{5}{6}$ of a meter and another piece that is $\frac{7}{12}$ of a meter. How much rope is left?

10. You have $\frac{3}{4}$ of a pizza, and your friend has $\frac{5}{8}$ of the same size pizza. If you combine them, how much pizza do you have altogether?

11. A car uses $\frac{3}{5}$ of a tank of gas on one trip and $\frac{2}{7}$ of a tank on another trip. How much of a tank of gas is used altogether?

12. A cyclist rides $12\frac{1}{2}$ miles on Monday and $8\frac{3}{4}$ miles on Tuesday. How many miles did the cyclist ride in total?

13. A juice recipe requires $\frac{2}{3}$ of a cup of orange juice and $\frac{3}{5}$ of a cup of pineapple juice. How many cups of juice are needed in total?

14. A box of nails weighs $\frac{7}{8}$ of a pound. If you buy 4 boxes, how many pounds of nails do you have?

15. A swimming pool is being filled at a rate of $\frac{2}{3}$ of a gallon per minute. How many gallons of water will be added in 15 minutes?

16. A gardener uses $\frac{3}{4}$ of a bag of soil for one planter and $\frac{5}{6}$ of a bag for another. How many bags of soil does the gardener use altogether?

17. You buy $\frac{2}{3}$ of a pound of apples and $\frac{3}{4}$ of a pound of bananas. The apples cost \$1.20 per pound, and the bananas cost \$0.80 per pound. What is the total cost?

18. A skateboarder rides $\frac{5}{6}$ of a mile in the morning, then twice that distance in the afternoon. How far does the skateboarder ride in total?

19. A painter mixes $\frac{3}{5}$ of a gallon of red paint with $\frac{7}{10}$ of a gallon of white paint. How many gallons of paint are mixed altogether?

20. A scientist has a solution that contains $\frac{7}{8}$ of a liter of chemical A and adds another $\frac{3}{4}$ of a liter of chemical B. What is the total volume of the solution?

21. A movie lasts $2\frac{1}{4}$ hours, and a documentary lasts $1\frac{2}{3}$ hours. How long would it take to watch both back-to-back?

22. A train travels $\frac{2}{3}$ of a mile each minute. How far will it travel in 45 minutes?

23. A recipe calls for $\frac{3}{4}$ of a cup of sugar. If you want to make $\frac{2}{5}$ of the recipe, how much sugar do you need?

1.3 Solving Equations

Objectives

In this section, we learn how to:

1. .
2. .
3. .

1.3.1 Applications

Intro:

1. Text
2. Text
3. Text

1.3.2 Introduction to Equations

We are going to introduce equations, which are not the same as expressions. Before introducing what an equation is, recall from Definition 1.1.2 that an expression is a mathematical notation representing a number. For example 23 is an expression and $\frac{7}{5}$ is an expression and $4x + 5$ is also an expression.

Definition 1.3.1

An **equation** is a mathematical statement that two expressions are equal. For example, $3x + 2 = 11$ is an equation stating that the expression $3x + 2$ is equal to 11

The key thing to note about *one* equation is that it states in its writing that *two* expressions are equal. An equation is saying to us "this" and "that" might not look the same, but "this" and "that" actually have the same value.

Example 1.3.2

Is $2x + 3 = 11$ an equation or an expression?

Solution. $2x + 3 = 11$ is an equation because it is stating that the expression $2x + 3$ is equal to the expression 11.

Example 1.3.3

Is $4y - 5$ an equation or an expression?

Solution. $4y-5$ is an expression because it is not stating that two expressions are equal. It is just one expression: the value of $4y - 5$ is one number, based on whatever the mystery value of y is.

Try it 1.3.4

Both expressions and equations are mathematical writing involving notation. How would you explain to someone how to spot the difference between an expression and an equation?

Example 1.3.5

Which of the following are equations and which are expressions?

1. $5x + 7 = 3$
2. $6y - 4$
3. $8 = 2z + 1$
4. $12 = 4 + x$
5. $9a + 5b$
6. $7 - 3c$

Note 1.3.6 The reason that it is important to know the difference between expressions and equations is that we will be doing different things with them.

1. For an expression, our primary task will be to *simplify* the expression we are given.
2. For an equation, our primary task will be to *solve* the equation we are given.

Warning 1.3.7 An equation is not the same as an expression. We cannot "solve" an expression. We can only "simplify" an expression. The action of *solving* will be language we reserve for equations.

The primary tool we have for solving equations is to apply the same action to both sides. To be more specific:

Apply the same action to both sides.

Given an equation, we can:

1. Add the same expression to both sides of the equation.
2. Subtract the same expression from both sides of the equation.
3. Multiply by the same expression on both sides of the equation.
4. Divide by the same non-zero expression on both sides of the equation.

1.3.3 Equations when the variable is written once

What we just described tells us what the typical steps we can do are. (There are some other things we can do, but let's wait to explain those until we need them.) The key thing to remember is that whatever we do to one side of the equation, we must do to the other side of the equation. Saying "do the same thing to both sides" tells us what we can do, but it doesn't really give us a lot of guidace on how to do it effectively. There are many kinds of equations we will see in this algebra class. I encourage you to look for patterns and ask "why does this work in this equation, but not in another equation?" That said, I think it's helpful to be completely transparent, and admit that there is a bit of an art to solving equations. Because of this, I want to share specific situations and explain specific strategies for these situations.

We are going to start by looking at equations where the variable is written once. Examples include equations like $x - 9 = 5$ and $7x + 2 = 29$. There are equations where the where the variable is written more than once such as $2x + 3 = 7x - 8$, and these will be discussed after.

Note 1.3.8 To begin, it will be helpful to slow down by writing in the writing the action we are applying to both sides as its own step, then simplifying both sides as a separate step. While this is more time-consuming (and we'll eventually stop doing it this way), skipping this and working too fast makes it impossible to describe certain common errors. To be able to point out what's going on, we need to start slow. I ask for your patience in doing it this way for now. Once the important points have been made, we'll make the process a little smoother.

Example 1.3.9

Solve the equation $x + 3 = 30$.

Solution. To undo the $+3$ on the left side of the equation, we can subtract 3 from both sides of the equation:

$$x + 3 = 30$$

$$x + 3 - 3 = 30 - 3$$
$$x = 27$$

On the left side, the $+3$ and the -3 together simplify to $+0$, so we are left with just x. On the right side, $30 - 3$ simplifies to 27.

In the answer to this first example, you might have felt annoyed that we wrote the equation $x + 3 - 3 = 30 - 3$ as its own step. Writing this equation is exactly what Note 1.3.8 is addressing. While you may feel like skipping this step in the future, let's just do a couple more where we write this step out.

Example 1.3.10

Solve the equation $x - 3 = 30$.

Solution. We can add 3 to both sides of the equation.

$$x - 3 = 30$$
$$x - 3 + 3 = 30 + 3$$
$$x = 33$$

Warning 1.3.11 Now, let's examine a new equation where a certain type of error tends to first occur from learners. Given the equation

$$x + 3 = 30$$

a common error is to end up at

$$x = 10.$$

Let's analyze this thorougly:

1. First, does the answer obtained $x = 10$ make sense? If we substitue 10 for x in the original equation, we get $10 + 3 = 30$, which is not true. So, $x = 10$ cannot be the correct answer.

2. While we already see that there is a problem, we haven't discussed how the error occurred. Let's try to figure that out. Here, it will be very helpful to write the action taken on both sides without simplifying right away, just as Note 1.3.8 talked about:

   $$x + 3 = 30$$

   From the original equation, based on the final answer that someone wrote, I would guess that they thought about dividing by 3 on both sides. Writing this without simplifying first looks like this:

$$\frac{x+3}{3} = \frac{30}{3}.$$

Here, it is true that the right side simplifies to 10. However, the expression on the left side says $\frac{x+3}{3}$ I would guess the learner "canceled" the 3 on top and bottom, but the 3 appearing in the numerator is a term, not a factor. Recall that we can only cancel factors, and not terms. It would be hard to analyze this error if we hadn't slowed the process down.

3. To stick to principles, we can divide by 3 on both sides. What happens is this:

$$\begin{aligned} x+3 &= 30 \\ \frac{x+3}{3} &= \frac{30}{3} \\ \frac{x+3}{3} &= 10 \end{aligned}$$

The steps shown above are all valid, but we haven't successfully solved for x, since x is not by itself on one side of the equation. This goes to show that while it is permissible to divide by 3 on both sides, it is not helpful in solving this equation. Let's show a couple more examples of sucessfully solving equations, and then summarize what kind of actions are helpful: we don't want to just "do" steps to do them; we want to find steps that "lead towards" our goal of isolating the variable on one side of the equation.

Example 1.3.12

Solve the equation $3x = 30$.

Solution. We can divide both sides of the equation by 3.

$$\begin{aligned} 3x &= 30 \\ \frac{3x}{3} &= \frac{30}{3} \\ x &= 10 \end{aligned}$$

In this example, dividing both sides by 3 was helpful. This is because in the fraction $\frac{3x}{3}$ there is a factor of 3 in both the numerator and denominator, so we can cancel the 3s. In slightly more detail, we can think of $\frac{3x}{3}$ as $\frac{3 \cdot x}{3 \cdot 1}$, which shows more clearly that 3 is a factor in both the numerator and denominator.

Note 1.3.13 In both of the last two examples, we divided by 3. In one example, this was helpful, and in the other example, this was not helpful. I want you to notice that when we divide both sides by a quantity, we end up creating a fraction even though we sometimes work so fast that we don't write the fraction that Note 1.3.8 encourages us to write. That fraction is there, whether we write it or are too lazy to write it, and we need to carefully apply what we learned in the last section about fractions: we can only cancel common factors.

Example 1.3.14

Solve the equation $\frac{x}{3} = 30$.

Solution. We can think of the left side as $x \div 3$. To undo the division by 3, we can multiply both sides by 3. We can multiply both sides of the equation by 3.

$$\begin{aligned} \frac{x}{3} &= 30 \\ \frac{x}{3} \cdot 3 &= 30 \cdot 3 \\ x &= 90 \end{aligned}$$

In this example, multiplying both sides by 3 was helpful. Why? Of the three equations that were one right after another, the left side of the second equation is the expression $\frac{x}{3} \cdot 3$. From the last section, we can treat the product of a fraction and a non-fraction in the usual way to get $\frac{x}{3} \cdot \frac{3}{1}$. Whether you multiply across first and cancel the factors of 3 on top and bottom after, or note that the 3s on top and bottom in the separate fractions would eventually become factors in the combined fraction $\frac{x \cdot 3}{3 \cdot 1}$ and cancel right away, we're left with $\frac{x}{1}$ which simplifies to x.

Solving Equations by Undoing Operations.

A general principle that is helpful to keep in mind is that we can think of solving equations as undoing operations. In more detail:

1. Subtraction undoes addition.

2. Addition undoes subtraction.

3. Division undoes multiplication.

4. Multiplication undoes division.

Warning 1.3.15 It is tempting to divide on both sides by 7 in an equation like $x - 7 = 31$ or $x - 7 = 2$. However, division does not undo subtraction. (Instead, addition undoes subtraction.)

Similarly, it is tempting to divide on both sides by 7 in an equation like $x + 7 = 40$ or $x + 7 = 5$. However, division does not undo addition. (Instead, subtraction undoes addition.)

In all four equations that have been mentioned in this box, dividing both sides by 7 is permissible, but not helpful. What ends up happening (if we slow things down the way Note 1.3.8 encourages us to) is that we have a fraction where the 7 in the numerator is a term, not a factor, so we cannot cancel it.

Example 1.3.16

Solve the equation $x + 30 = 2$.

Solution 1.

$$\begin{aligned} x + 30 &= 2 \\ x + 30 - 30 &= 2 - 30 \\ x &= -28 \end{aligned}$$

Solution 2. Since we made the point about writing the action that occurs on both sides without simplyfing in Note 1.3.8 so that we could analyze a situation like we did in Warning 1.3.11, we can now speed things up a little and show a perfectly acceptable answer to this question which doesn't show the middles step.

$$\begin{aligned} x + 30 &= 2 \\ x &= -28 \end{aligned}$$

While both answers are fine, we will still have situations later in the book where writing the action but not simplifying yet (as described in Note 1.3.8) makes it possible for us to discuss what went wrong. The overall point here is not to criticize skipping steps. Instead, if someone's work goes from Equation 1 to Equation 3 while skipping the writing of Equation 2, but there was an error going from Equation 2 to Equation 3, then we need to actually slow down by writing in all the steps. Skipping work is fine when the work is correct. But if there is an error hiding somewhere inside skipped work, then it is really helpful to include all the steps to be able to pinpoint exactly where the error occured.

Example 1.3.17

Solve the equation $2x = 25$.

Solution 1.

$$\begin{aligned} 2x &= 25 \\ \frac{2x}{2} &= \frac{25}{2} \\ x &= \frac{25}{2} \end{aligned}$$

Solution 2. Let's also present a solution which skips showing the action, and instead shows the result of simplifying where possible.

$$\begin{aligned} 2x &= 25 \\ x &= \frac{25}{2} \end{aligned}$$

While the two answers showed a different amount of work (by skipping the writing of one equation), we were able to cancel the 2 on top and on bottom (since 2 was a factor on top and a factor on bottom), even if we didn't explicitly write the fraction in the second answer. In both answers, $\frac{25}{2}$ didn't simplify further. If an application called for it, we might present this number as a mixed fraction or a decimal, but recall that leaving $\frac{25}{2}$ as written is what we usually prefer in algebra.

Example 1.3.18

Solve the equation $8x = 5$.

Solution.

$$\begin{aligned} 8x &= 5 \\ x &= \frac{5}{8} \end{aligned}$$

Example 1.3.19

Solve the equation $3 = x - 20$.

Solution.

$$\begin{aligned} 3 &= x - 20 \\ 23 &= x \end{aligned}$$

This last example shows that it's okay to have the variable on the right side of the final equation.

Equations with several operations

So far, we have only seen equations where a single operation is being applied to the variable, and therfore we applied an action to both sides and simplified, and we were done. We only needed to apply one action to both sides.

Let's now look at a more complicated equation. For example, consider the equation $3x + 2 = 17$. While it may seems strange to you, I want to first point out a common error that learners make when solving this equation. Sometimes, as a first step, people will divide both sides by 3 and might not write the step in between that is discussed in Note 1.3.8, but we will:

$$\frac{3x+2}{3} = \frac{17}{3}$$

What usually happens is that people will skip writing this step, and then they will "cancel" the 3s on top and bottom, getting $x + 2 = \frac{17}{3}$. However, this is an error, because the 3 in the numerator is not a factor of the numerator. (Instead, the numerator has a term of $3x$ and a term of 2.) Divide the *entire* left and entire right sides of the equation by the same non-zero value.

1.3.4 AAAAA

Technique with one copy of the variable. How to organize work for solving equations. How the work for solving equations looks different for the work of simplifying expressions, and why it matters. What solving an equation MEANS. When are we done: need ONE copy of the variable on one side, by itself Left versus right side? Multiple terms. Rewriting as multiple terms. Strategy comparison.

text.

text

Example 1.3.20

Statement text

Solution. Solution text

Try it 1.3.21

Text

Note 1.3.22 Text

Title.

Text

Warning 1.3.23 text

1.3.5 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section.

Example 1.3.24

Text

Solution. Answer

1.3.6 Summary

1. Summary point.
2. Summary point.
3. Summary point.

1.3.7 Exercises

1. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart
2. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart
3. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart

1.4 Equations with Fractions

Text of section.

1.5 Exponents

Text of section.

1.6 Radicals

Text of section.

1.7 Factoring and Expansion

Text of section. Systematize the factoring of quadratic trinomials. Add warning that we shouldn't set an expression equal to zero and create an equation to solve: gently preview what the student is confusing this with

1.8 Rational Expressions

Text of section.

1.9 Solving Equations Revisited

Text of section. More applications, more challenging (up to linear) equations, and fractional equations (checking denominator). Applications Literal equations (multiple variables) Systems of equations Justifying cross-multiplication

1.10 Sample code

Objectives

After completing this section, you should be able to do the following.

1. Explain the conditions under which an implication is true.
2. Identify statements as equivalent to a given implication or its converse.
3. Explain the relationship between the truth values of an implication, its converse, and its contrapositive.

1.10.1 Section Preview

Investigate!

While walking through a fictional forest, you encounter three trolls guarding a bridge. Each is either a *knight*, who always tells the truth, or a *knave*, who always lies. The trolls will not let you pass until you correctly identify each as either a knight or a knave. Each troll makes a single statement:

Troll 1: If I am a knave, then there are exactly two knights here.

Troll 2: Troll 1 is lying.

Troll 3: Either we are all knaves, or at least one of us is a knight.

Which troll is which?

Try it 1.10.1

Spend a few minutes thinking about the Investigate problem above. What could you conclude if you knew Troll 1 really was a knave (i.e., their statement was false)? Share your initial thoughts on this.

Definition 1.10.2 Argument.

An **argument** is a sequence of statements, the last of which is called the **conclusion** and the rest of which are called **premises**.

An argument is said to be **valid** provided the conclusion must be true whenever the premises are all true. An argument is **invalid** if it is not valid; that is, all the premises can be true, and the conclusion could still be false.

An argument is **sound** provided it is valid and all the premises are true. A

proof of a statement is a sound argument whose conclusion is the statement.

By the way... Our definitions of **argument**, **valid argument**, and **sound argument** are the same ones used in philosophy, the other primary academic discipline concerned with logic and reasoning.

Example 1.10.3

Consider the following two arguments:

$$\begin{array}{rl} & \text{If Edith eats her vegetables, then she can have a cookie.} \\ & \text{Edith eats her vegetables.} \\ \hline \therefore & \text{Edith gets a cookie.} \end{array}$$

$$\begin{array}{rl} & \text{Florence must eat her vegetables to get a cookie.} \\ & \text{Florence eats her vegetables.} \\ \hline \therefore & \text{Florence gets a cookie.} \end{array}$$

(The symbol "$\therefore$" means "therefore")

Are these arguments valid?

Solution. Do you agree that the first argument is valid but the second argument is not? We will soon develop a better understanding of the logic involved in this analysis, but if your intuition agrees with this assessment, then you are in good shape.

Notice the two arguments look almost identical. Edith and Florence both eat their vegetables. In both cases, there is a connection between the eating of vegetables and cookies. Yet we claim that it is valid to conclude that Edith gets a cookie, but not that Florence does. The difference must be in the connection between eating vegetables and getting cookies. We need to be skilled at reading and comprehending these sentences. Do the two sentences mean the same thing?

Unfortunately, in everyday language we are often sloppy, and you might be tempted to say they are equivalent. But notice that just because Florence *must* eat her vegetables, we have not claimed that doing so would be *enough* (she might also need to clean her room, for example). In everyday (non-mathematical) practice, you might be tempted to say this "other direction" is implied. In mathematics, we never get that luxury.

Remark 1.10.4 The arguments in the example above illustrate another important point: Even if you don't care about the advancement of human knowledge in the field of mathematics, becoming skilled at analyzing arguments is useful. And even if you don't want to give your grandmother a cookie. If you are *using* mathematics to solve problems in some other discipline, it is still necessary to demonstrate that your solution is correct. You better have a good argument that it is!

Try it 1.10.5

Test

To show why this warning is here, in Try it 1.10.5